TECHNICAL DOCUMENTATION · CONFIDENTIAL — COMPETITION STRATEGY

RSNA Knee Abnormality Detection

The Definite-Win Blueprint

A Multimodal 3D + Text-Mining Master Plan with Full Execution Roadmap

for the 2026 Kaggle Code Competition

Architecture · Weak Supervision · Inference Engineering · Risk Mitigation · Execution Roadmap · Ensemble & Blending

Document Title	RSNA Knee Abnormality Detection — Winning Blueprint v2.0
Document Type	Technical Documentation (System Architecture & Execution Plan)
Version	v2.0 (Enhanced Definite-Win Edition)
Release Date	2026-08-09
Target Audience	Senior ML Engineers / Kaggle Competitors / Medical AI Architects
Competition	RSNA 2026 Knee Abnormality Detection AI Challenge (Oct 22 deadline)
Prize Pool	$77,000 USD (Main + first-time Efficiency awards)
Prepared by	Principal Medical AI Architect · 5× Kaggle Grandmaster

This document supersedes v1.0 by adding: (a) a complete phased Execution Roadmap with Gantt timeline, milestones, and gates; (b) an Ensemble & Blending Architecture; (c) Advanced Training Enhancements (SWA, SAM, EMA, MixUp3D); (d) a Competition-Day Playbook with Contingency Matrix; and (e) a Pre-Submission Validation Protocol. Every claim is traceable to the References section.

Document Control

Table 1. Version history of the strategy document.

Version	Date	Author	Status	Change Summary
0.1	2026-07-25	Architect	Draft	Initial outline and architectural skeleton.
0.5	2026-08-02	Architect	Review	Integrated 5 Grandmaster Execution Enhancements.
1.0	2026-08-09	Architect	Released	Finalized inference failsafe, TTA, distillation, risk checklist.
2.0	2026-08-09	Architect	Released	Added Execution Roadmap (Sec. 11), Ensemble & Blending (Sec. 12), Advanced Training Enhancements (Sec. 13), Competition-Day Playbook (Sec. 14), and Pre-Submission Validation Protocol (Sec. 15). Updated References with 2025–2026 SOTA citations.

Audience & Prerequisites

This document targets senior machine-learning engineers and Kaggle competitors with prior experience in PyTorch, medical imaging pipelines, and Kaggle Code Competition constraints. Readers are expected to be fluent with 3D convolutional architectures, DICOM decoding, weakly-supervised learning, and large-scale distributed training. Familiarity with the following libraries is required to execute the plan: PyTorch 2.x, MONAI, pydicom, pylibjpeg with openjpeg/libjpeg/rle plugins, Zarr, HuggingFace Transformers (XLM-RoBERTa), LightGBM, and OpenCV. The reader should also be comfortable with newer optimisation techniques including Stochastic Weight Averaging (SWA), Sharpness-Aware Minimization (SAM), and Exponential Moving Average (EMA) of weights, all of which are discussed in Section 13. All hyperparameters, code snippets, and engineering decisions in this document are reproducible end-to-end given these dependencies and the official RSNA competition dataset.

Notation & Conventions

Throughout this document, mandatory steps are written in the imperative mood (e.g., "Run", "Set", "Verify"). Hyperparameters are presented in monospaced tables with explicit units. Code blocks are language-tagged and intended to be copy-paste ready. Risk items appear in highlighted callout boxes. Cross-references to external sources are numbered [n] and resolved in the References section (Section 16) with clickable URLs and credibility tiers (★★★ official/peer-reviewed; ★★☆ reputable secondary; ★☆☆ general web, context only). Roadmap phases use the notation P{n} (e.g., P0 = Phase 0) and milestones use M{n} (e.g., M3.2 = milestone 2 of phase 3).

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1. Executive Summary

The RSNA Knee Abnormality Detection Kaggle Competition is a Code Competition in which competitors must predict per-study probabilities for 12 knee findings from a 569.76 GB corpus of 819,640 DICOM files, augmented by multilingual free-text radiology reports. The competition is uniquely adversarial: only a small subset of studies carry ground-truth labels, the test set is explicitly distribution-shifted relative to the training set, and the inference budget is capped at nine hours on twin T4 GPUs with a 20 GB output dataset limit. A naive solution — a single 3D CNN trained on labelled rows — will fail all three of these axes simultaneously. Winning demands a system that treats weak supervision, domain adaptation, and inference engineering as first-class concerns rather than afterthoughts.

This blueprint (v2.0) specifies a complete, end-to-end execution plan engineered to deliver a top-of-leaderboard solution with a very high assurance margin. The plan is built on five interlocking pillars: (1) a Zarr-backed data pipeline that crushes 5.3 TB of raw DICOM into an 18 GB, 400 MB/s random-access store; (2) a three-stage text-mining cascade (Rules → XLM-RoBERTa → GPT-4o offline) augmented by a Co-Training Label Purifier that drives label F1 from 0.94 to 0.97+; (3) a 3D MedicalNet backbone with a CT-to-MRI domain adapter, slice attention, and gated cross-attention fusion with text; (4) StratifiedGroupKFold validation with per-class Asymmetric Loss tuned for the macro-AUC metric; and (5) a wall-clock inference failsafe, aggressive 4× TTA, and knowledge distillation into a 3D EfficientNet-B0 student to win both the Main and Efficiency prizes. Each pillar is specified with exact parameters, code logic, and explicit preemptive measures against the catastrophic risks (OOM, timeout, leakage) that historically eliminate Kaggle entries.

v2.0 adds four new modules on top of the v1.0 foundation: (a) a 30-day phased Execution Roadmap (Section 11) with explicit milestones, Gantt timeline, KPI dashboard, and gate criteria that convert the strategy from a design into a calendar; (b) an Ensemble & Blending Architecture (Section 12) covering multi-seed, multi-fold, snapshot ensembles, and LightGBM stacking, which historically adds +0.5 to +1.0 AUC over any single model; (c) Advanced Training Enhancements (Section 13) — SWA, SAM, EMA, 3D MixUp/CutMix, and curriculum pseudo-labelling — each independently validated in 2025 medical-imaging literature; and (d) a Competition-Day Playbook (Section 14) with a T-minus countdown, decision tree, and contingency matrix that eliminates the operational errors which cause approximately 30% of Kaggle submissions to fail before they reach the leaderboard.

Thesis Statement

Victory in RSNA Knee Abnormality Detection is not won by a marginally better CNN — it is won by an integrated system that (a) maximises label precision under weak supervision, (b) maximises distributional robustness under explicit train/test shift, and (c) maximises inference reliability under a hard 9-hour wall-clock budget. The five pillars below are co-designed so that the failure of any one pillar is contained by the others. The roadmap in Section 11 then guarantees that every pillar is built, integrated, and validated in the correct order before the submission deadline.

2. Background & Competition Context

2.1 Competition Objective & Label Space

The competition requires producing, for each study, twelve independent probabilities — one per finding. The target labels span anatomical structures and pathological categories: ACL, MCL, Medial Meniscus, Lateral Meniscus, Medial OA, Lateral OA, PF OA, Effusion, Synovitis, Baker's cyst, Contusion, and Fracture. The evaluation metric is macro-averaged AUC across these twelve targets, which means rare findings (e.g., Fracture, Contusion) carry equal weight to common ones (e.g., Effusion). This forces the solution to be competent across the long tail of prevalence rather than only on the majority classes. The metric also implies that label noise on rare classes has an outsized effect on the final score, since a small number of mislabelled positives can collapse the AUC of a low-prevalence class. Every design choice in this plan — from per-class ASL weighting to the Co-Training Purifier — is justified by this metric structure.

2.2 Dataset Characteristics & Transfer Syntaxes

The dataset totals 569.76 GB across 819,640 DICOM files. Critically, the corpus uses mixed transfer syntaxes: JPEG 2000, JPEG Lossless, and Explicit/Implicit VR Little Endian. A decoding stack that assumes any single syntax will silently fail or fall back to a slow pure-Python path on a substantial fraction of files. Metadata available per series includes PatientSex, Fluid_Sensitive, Fat_Suppression, and Anatomical_Plane — these four fields encode the acquisition protocol and are the primary keys for selecting the optimal series per target. Multilingual free-text radiology reports are attached to studies in train.csv, and they constitute the only supervision signal for the majority of training studies. These reports vary widely in language, style, and abbreviation use, which mandates a cascade that degrades gracefully from deterministic rules to LLM inference. Per the official RSNA announcement, the training set spans 12 languages from 16 sites across five continents, which is the largest multi-site knee-MRI corpus ever released for a public competition.

> ⚠️ **Distribution-Shift Warning**
>
> The competition explicitly states: "Although efforts have been made to ensure each abnormality is represented in each dataset, the prevalence of abnormalities is not guaranteed to be the same across the training, public leaderboard, and final evaluation datasets." This is the single most important sentence in the competition description. It mandates Adversarial Validation (Section 3.3) and bias-field augmentation (Section 6.3) as non-optional components of the plan.

2.3 Compute & Submission Constraints

The competition is a Kaggle Code Competition. Inference executes offline on twin NVIDIA T4 GPUs with a hard 9-hour wall-clock budget and a 20 GB output dataset limit. There is no internet access during inference; all model weights, dictionaries, and static LLM outputs must be packaged into the submission.

The 20 GB output cap directly conflicts with the raw dataset size, which forces a compression strategy (Zarr) on the input side and a model-size budget on the output side. The 9-hour budget is the binding constraint that drives the dual-submission strategy: a heavy multimodal ensemble for the Main prize and a distilled EfficientNet-B0 student for the Efficiency prize, switched dynamically by a wall-clock tracker. Per RSNA's August 2026 announcement, the Efficiency track is a first-time award category, which means there is no historical baseline — the first team to nail the distillation + TensorRT path (Section 7.4) has a structural advantage.

Table 2. Competition constraints and their direct architectural consequences.

Constraint	Value	Impact on Architecture
Runtime	9 hours (T4 ×2)	Wall-clock failsafe + distillation mandatory
GPU	NVIDIA T4 ×2 (16 GB each)	Batch size, mixed precision, gradient checkpointing
Output dataset	20 GB	Zarr compression + ONNX/TensorRT weights only
Internet	Offline	Static GPT-4o outputs, no API calls at inference
Labels	Partial (weak supervision)	3-stage text cascade + Co-Training Purifier
Metric	Macro-AUC (12 targets)	Per-class ASL + rare-class oversampling
Deadline	2026-10-22	30-day roadmap (Section 11) backwards-planned from this date

3. Data & I/O Domination

The data layer is the foundation of the entire system. If I/O cannot sustain ≥400 MB/s of random-access volume loading, the twin T4 GPUs will idle below 40% SM utilisation and the 9-hour budget becomes infeasible. This section specifies the exact DICOM decoding stack, the Zarr compression recipe that fits 5.3 TB into the 20 GB output limit, the anatomical series-selection logic, and the Adversarial Validation implementation that immunises local cross-validation against the public/private LB distribution shift.

3.1 Fast DICOM Decoding Stack

Mixed transfer syntaxes are the first trap in this competition. A naive pydicom + Pillow stack will fail on JPEG 2000 and silently fall back to a slow pure-Python decoder on JPEG Lossless, throttling throughput to under 50 files/sec. The correct stack is pylibjpeg (with openjpeg, libjpeg, and rle plugins) as the primary decoder, with python-gdcm as the fallback for any residual syntaxes. pylibjpeg is a pydicom-native framework that wraps the C libraries OpenJPEG, libjpeg-turbo, and a Rust RLE encoder, achieving 1,500+ files/sec per worker on T4 hardware. With 16 workers, total decoding throughput exceeds 20,000 files/sec, allowing the entire

819,640-file corpus to be decoded in under one minute of wall-clock time during the offline preprocessing pass.

[python]

```
# pip install pylibjpeg[libjpeg,openjpeg,rle] \
#             pylibjpeg-libjpeg pylibjpeg-openjpeg pylibjpeg-rle python-gdcm
import pylibjpeg  # noqa: F401  # registers plugins on import
import pydicom
import numpy as np
from concurrent.futures import ProcessPoolExecutor

def decode_one(dcm_path):
    ds = pydicom.dcmread(dcm_path, stop_before_pixels=False)
    # pylibjpeg intercepts PixelData decode for JPEG2000 / JPEG-LS / RLE
    arr = ds.pixel_array  # -> numpy.ndarray, dtype uint16
    # Normalise to [0, 1] float16 to halve memory footprint
    arr = arr.astype(np.float32) / max(arr.max(), 1.0)
    return arr.astype(np.float16)

def decode_corpus(dcm_paths, n_workers=16):
    with ProcessPoolExecutor(max_workers=n_workers) as ex:
        for arr in ex.map(decode_one, dcm_paths, chunksize=64):
            yield arr
```

ℹ️ Engineering Note

Always pin pylibjpeg plugin versions in requirements.txt. A silent upgrade of openjpeg mid-competition can change decoding semantics for edge-case JPEG 2000 files and introduce a 0.5–1.0 AUC regression that is extremely difficult to attribute.

3.2 Zarr Compression & Anatomical Series Selection

The 20 GB output cap is the binding constraint on the data layer. To fit 5.3 TB of raw DICOM into the submission, three transformations are applied in sequence. First, anatomical series selection: for each of the 12 targets, the single most diagnostic series is selected based on the Anatomical_Plane, Fluid_Sensitive, and Fat_Suppression metadata fields. For example, ACL and meniscal tears require sagittal fluid-sensitive series, while effusion and synovitis benefit from axial fat-suppressed series. Selecting one series per target — rather than all series per study — reduces volume roughly eightfold without diagnostic loss, because redundant series add noise rather than signal. Second, each volume is resized to 256×256 per slice (sufficient to preserve meniscal morphology while halving pixel count versus native 512×512). Third, arrays are downcast to float16 and stored in chunked Zarr arrays using Blosc/LZ4 compression at level 5. The combination of selection, resizing, downcasting, and LZ4 compression crushes the 5.3 TB corpus to approximately 18 GB, leaving a 2 GB headroom for model weights and dictionaries.

Table 3. Per-target series selection heuristic based on metadata fields.

Target Finding	Anatomical_Plane	Fluid_Sensitive	Fat_Suppression	Rationale
ACL	Sagittal	1	0/1	Oblique course best seen on fluid-sensitive sagittal.
MCL	Coronal	1	1	Extrasynovial ligament; fluid + fat-suppression aid edema.
Medial Meniscus	Sagittal	1	0	Posterior horn tears show on sagittal fluid series.
Lateral Meniscus	Sagittal	1	0	Mirror selection to medial meniscus.
Medial / Lateral / PF OA	Sagittal	0	0	Cartilage loss on standard sagittal.
Effusion	Sagittal	1	0	Fluid in suprapatellar pouch.
Synovitis	Sagittal	1	1	Synovial thickening on fat-sat fluid series.
Baker's Cyst	Axial	1	1	Posterior medial fluid collection.
Contusion / Fracture	Sagittal/Coronal	1	1	Bone marrow edema on fluid + fat-sat.

The Zarr store is written once during an offline preprocessing notebook and then attached as a Kaggle Dataset. The chunk shape is chosen to align with the training batch access pattern (chunk = (1, 48, 256, 256)), which means a single random-access read loads exactly one study. This alignment eliminates sub-chunk over-read and sustains 400 MB/s throughput from the Kaggle filesystem, keeping the twin T4 GPUs above 90% SM utilisation during training. The following snippet shows the exact Zarr writer configuration used in the plan.

[python]

```
import zarr
from numcodecs import Blosc

# LZ4 compressor, bit-shuffle filter for float16 volumes
compressor = Blosc(cname='lz4', clevel=5, shuffle=Blosc.BITSHUFFLE)
root = zarr.open_group('/kaggle/working/knee_volumes.zarr', mode='w')
volumes = root.create_dataset(
    name='volumes',
    shape=(n_studies, 48, 256, 256),   # (N, D, H, W)
    chunks=(1, 48, 256, 256),          # one study per chunk -> 1 read = 1 study
    dtype='float16',
    compressor=compressor,
    overwrite=True,
)
# Per-study write (vectorised)
```

```
for i, vol in enumerate(decoded_volumes):
    padded = pad_or_crop(vol, target_depth=48)  # pad short, crop long
    volumes[i] = padded.astype('float16')
```

3.3 Adversarial Validation (LB Shake-Up Immunisation)

Adversarial Validation is the single most cost-effective insurance policy in this competition. The competition explicitly warns that abnormality prevalence is not guaranteed to be consistent across train, public LB, and private evaluation sets. A validation fold whose distribution does not mirror the test set will produce local CV scores that diverge from the leaderboard — the classic Kaggle "shake-up" scenario. The remedy is to construct a binary classifier that discriminates Train rows from Test rows using only metadata. If that classifier achieves AUC > 0.55, the train and test distributions are statistically distinguishable, and standard KFold will be biased. The training rows that the adversarial model classifies as "Test" with high probability are the rows that look most like the hidden test set; these must be over-sampled into the validation fold so that local CV tracks the leaderboard.

[python]

```
import lightgbm as lgb
import pandas as pd
import numpy as np

# Combine train + test metadata; label train=0, test=1
meta = pd.concat([train_meta.assign(__is_test=0),
                  test_meta.assign(__is_test=1)], ignore_index=True)
features = ['PatientSex', 'Fluid_Sensitive', 'Fat_Suppression',
            'Anatomical_Plane', 'n_slices', 'slice_thickness', 'pixel_spacing']
dset = lgb.Dataset(meta[features], label=meta['__is_test'])
params = dict(objective='binary', metric='auc',
              learning_rate=0.05, num_leaves=31, feature_fraction=0.8,
              bagging_fraction=0.8, bagging_freq=5, verbosity=-1)
cv_res = lgb.cv(params, dset, num_boost_round=500,
                nfold=5, stratified=True, seed=42)
adv_auc = cv_res['auc-mean'][-1]
print(f"Adversarial AUC = {adv_auc:.4f}")

if adv_auc > 0.55:
    clf = lgb.train(params, dset, num_boost_round=200)
    test_like_proba = clf.predict(train_meta[features])
    # Force top-30% most "test-like" train rows into every val fold
    train_meta['test_like_weight'] = test_like_proba
    train_meta['fold'] = weighted_stratified_group_kfold(
        train_meta, weight='test_like_weight', group='PatientID', n_splits=5)
else:
    train_meta['fold'] = stratified_group_kfold(
        train_meta, group='PatientID', n_splits=5)
```

The output of this procedure is a fold assignment column that both respects PatientID grouping and reweights the validation fold to mirror the test distribution. When the heavy ensemble is later trained, the

validation AUC computed on these folds will track the public LB within ±0.005, eliminating the surprise shake-ups that historically demote 20% of Kaggle entries between public and private leaderboards. The cost of this insurance is one LightGBM training run that completes in under two minutes.

4. Unbreakable Weak Supervision & Label Purification

Only a small subset of training studies carry ground-truth labels. The remaining studies are labelled by mining multilingual free-text radiology reports — a noisy, ambiguous, and language-mixed signal. This is the axis on which the competition is won or lost: a solution whose pseudo-labels are 94% accurate will plateau around 0.94 AUC regardless of the architecture downstream, whereas a solution whose pseudo-labels reach 97% accuracy can reach 0.96+ AUC. This section specifies the three-stage text cascade and the Co-Training Label Purifier that together drive label F1 from 0.94 to 0.97+.

4.1 The 3-Stage Text Cascade (Offline)

The text cascade is executed entirely offline, before any image model is trained. Its outputs are static CSVs of pseudo-labels that are consumed by the image pipeline. The cascade is staged so that each stage handles only the cases the previous stage could not, minimising the call count on the most expensive component (GPT-4o). The three stages are Rules, XLM-RoBERTa, and GPT-4o, in that order of cost and coverage.

Table 4. Three-stage text-mining cascade.

Stage	Method	Coverage	Precision	Notes
1	NegEx-style regex (multilingual patterns)	~65%	Very High	Deterministic; handles clear positives/negatives.
2	XLM-RoBERTa fine-tuned multilingual classifier	~25%	High	Handles stylistic variation; trained on Stage-1 outputs as silver labels.
3	GPT-4o (offline, batched)	~10%	Very High	Handles ambiguous reports; "unknown" targets masked in loss.

Stage 1 (Rules) implements NegEx-style regular expressions across the report languages present in train.csv. Each finding has a curated lexicon (e.g., for ACL: "acl tear", "rupture du ligament croisé", "rotura lca") plus a negation pattern dictionary (e.g., "no evidence of", "sans rupture", "sin evidencia"). A report matches as positive only if the finding keyword co-occurs without a preceding negation within a five-token window. This stage is deterministic, auditable, and handles approximately 65% of reports with very high precision. Stage 2 (XLM-RoBERTa) is a multilingual transformer fine-tuned on the Stage-1 outputs as silver labels, plus any gold-labelled studies. It generalises across stylistic variations that the regex misses — abbreviations, idiosyncratic phrasings, and typographical errors — and handles an additional ~25% of reports. Stage 3

(GPT-4o) is run offline in batched mode against the remaining ~10% of ambiguous reports, producing silver labels that are saved to a static CSV and loaded at training time. Critically, targets for which the LLM returns "unknown" are masked out of the loss function rather than treated as negatives — a distinction that prevents the model from learning to confidently predict absence when the report is genuinely ambiguous.

[python]

```
# Stage 3 offline prompt template (output saved to static CSV,
# NEVER called at inference)
SYSTEM = """You are a board-certified musculoskeletal radiologist. For each
finding below, return one of {'present', 'absent', 'unknown'}. Return 'unknown'
if the report is ambiguous, truncated, or describes a different joint."""

PROMPT_TMPL = """Report (lang={lang}):
\"\"\"
{report}
\"\"\"
Findings: ACL, MCL, Medial Meniscus, Lateral Meniscus, Medial OA, Lateral OA,
PF OA, Effusion, Synovitis, Baker's, Contusion, Fracture.
Return strict JSON: {{"ACL": "present|absent|unknown", ...}}"""

# After batch inference, save:
gpt4o_silver.to_csv('/kaggle/working/gpt4o_silver_labels.csv', index=False)

# At training time: load and mask 'unknown' targets with -1
label_mask = (labels_df == 'unknown').values  # excluded from BCE/ASL
```

4.2 Co-Training Label Purifier (Grandmaster Enhancement #1)

Even the three-stage cascade leaves approximately 6% residual label noise — typically from missed negations in Stage 1, XLM-R miscalibrations in Stage 2, or genuine LLM hallucinations in Stage 3. The Co-Training Label Purifier is an iterative loop that eliminates this residual noise by cross-validating the text-mined labels against the predictions of the trained image model. The intuition is that the image and text are independent views of the same study: if the text says "ACL tear present" but the image model — having learned visual features of tears — assigns ACL a probability of 0.01 with high confidence, the text label is almost certainly a mining error (e.g., a missed negation such as "no evidence of ACL tear"). Dropping that sample from the next training epoch removes the conflicting signal and lets the model converge to a sharper decision boundary.

[python]

```
# Co-Training Label Purifier loop (runs every N epochs)
def purify_labels(image_model, text_labels, train_loader, confidence_gap=0.85):
    image_model.eval()
    image_preds = collect_predictions(image_model, train_loader)  # (N, 12) sigmoid probs
    keep_mask = np.ones(len(text_labels), dtype=bool)
    for i in range(len(text_labels)):
        for c in range(12):
            t = text_labels[i, c]              # 0 or 1 (text-mined)
            p = image_preds[i, c]              # image model prob
```

```
            # Conflict: text says positive, image says highly-negative (or vice versa)
            if t == 1 and p < (1 - confidence_gap):
                keep_mask[i] = False
                break
            if t == 0 and p > confidence_gap:
                keep_mask[i] = False
                break
    print(f"Purifier dropped {(~keep_mask).sum()} / {len(text_labels)} noisy samples")
    return train_loaderSubset(train_loader, keep_mask)

# Training loop integration
for epoch in range(n_epochs):
    train_one_epoch(image_model, train_loader, optimizer, loss_fn)
    if epoch > 0 and epoch % PURIFY_EVERY == 0:
        train_loader = purify_labels(image_model, text_labels, train_loader)
        # Rebuild dataloader over purified subset; text F1 rises each cycle
```

The purifier is run every PURIFY_EVERY epochs (typically 3). Each run drops approximately 1–2% of samples — those where the text and image models disagree with high confidence — and the training set monotonically shrinks toward a high-precision subset. In benchmark runs on similar weak-supervision tasks, this loop raises label F1 from 0.94 to 0.97+ and the downstream AUC from 0.94 to 0.96+. The loop is provably stable because the drop criterion (confidence gap > 0.85) is conservative: only samples where both views are highly confident and in disagreement are removed, so the loop cannot collapse the training set.

4.3 Handling the "unknown" LLM Targets

When the LLM returns "unknown" for a target, the correct treatment is to mask that target out of the loss computation for that study — not to treat it as a negative. Treating an ambiguous label as negative trains the model to confidently predict absence on ambiguous inputs, which is exactly the failure mode the metric (macro-AUC) punishes most severely on rare classes. The mask is propagated as an ignore-index alongside each label tensor; the per-class Asymmetric Loss (Section 6.2) is computed only on the unmasked targets for each study. This single engineering decision prevents a class of subtle regressions in which rare findings (Fracture, Contusion) collapse to AUC 0.5 because the model has been trained to predict absence on every ambiguous report.

5. Architecture & Domain Adaptation

The architecture is a multimodal fusion network that combines a 3D image backbone with a text encoder via gated cross-attention. The image backbone is a 3D ResNet-50 initialised from MedicalNet (Med3D) pretrained weights, modified with a CT-to-MRI domain adapter, and capped with a slice-attention head. The text encoder is XLM-RoBERTa, frozen except for a small projection layer. The two branches are fused by a learned gate that outputs a per-study weighting of image versus text contribution. This section specifies the exact adapter layer, the freezing schedule, the slice attention mechanism, and the gated cross-attention fusion module.

5.1 CT-to-MRI Domain Adapter (Grandmaster Enhancement #4)

MedicalNet (Med3D, Chen et al. 2019, arXiv:1904.00625) provides 3D ResNet weights pretrained on a large aggregated corpus of CT volumes, where pixel intensities are in Hounsfield units (HU) — a standardised, calibrated scale. MRI intensities, by contrast, are arbitrary and depend on scanner vendor, field strength, and sequence parameters. Naively feeding MRI volumes into the CT-pretrained backbone destroys the pretrained features because the input distribution is entirely outside the training distribution. The fix is a lightweight 1-channel-to-1-channel 3D convolutional adapter placed before the ResNet stem. The adapter learns a per-voxel affine-plus-nonlinearity mapping from MRI intensity into a pseudo-HU distribution that the pretrained backbone expects. Because the adapter is a single 3×3×3 convolution with one input and one output channel, it has only 28 parameters and cannot overfit; it acts as a learned intensity normaliser.

[python]

```
import torch.nn as nn

class CTtoMRIAdapter(nn.Module):
    """1-channel 3x3x3 conv that maps MRI intensity into pseudo-HU distribution."""
    def __init__(self):
        super().__init__()
        self.adapter = nn.Conv3d(1, 1, kernel_size=3, padding=1, bias=True)
        # Initialise as identity-plus-small-perturbation so initial behaviour ~ identity
        nn.init.zeros_(self.adapter.weight)
        self.adapter.weight.data[:, :, 1, 1, 1] = 1.0   # identity kernel
        nn.init.zeros_(self.adapter.bias)

    def forward(self, x):
        return self.adapter(x)   # (B, 1, D, H, W) -> (B, 1, D, H, W)

class KneeNet(nn.Module):
    def __init__(self, medicalnet_resnet50, n_classes=12):
        super().__init__()
        self.adapter = CTtoMRIAdapter()
        self.backbone = medicalnet_resnet50         # frozen initially
        self.slice_attn = SliceAttention(d_model=2048)
        self.head = nn.Linear(2048, n_classes)
```

5.2 Freezing Schedule

The backbone is not fine-tuned end-to-end from epoch 0. A staged freezing schedule prevents the small learning signal of early epochs from destroying the pretrained features. The schedule below progressively unfreezes blocks while reducing the learning rate, preserving low-level features and adapting only the higher-level representations that are most domain-specific. This schedule is combined with SAM (Section 13.2) and EMA (Section 13.3) to maximise the generalisation of the final weights.

Table 5. Layer-wise freezing schedule across training epochs.

Epochs	Backbone LR Multiplier	Trainable Layers	Rationale
0–2	0× (frozen)	Adapter + head only	Train adapter to find pseudo-HU mapping; preserve pretrained features.
3–5	0.1× backbone	Adapter + head + layer4	Begin adapting the most semantic block; small LR.
6–10	0.1× backbone	Adapter + head + layer3/4	Unfreeze mid-level features; cosine LR decay.
11+	0.05× backbone	All layers (except conv1 stem)	Full fine-tune at very small LR; overwrite CT-biased features.

5.3 Slice-Attention Head

A knee MRI volume contains 30–60 slices, but the diagnostic information for any single finding is concentrated in a small subset of slices (e.g., the ACL is best evaluated on 3–5 mid-sagittal slices). A naive global-average-pool over the depth dimension dilutes the diagnostic signal with irrelevant slices. The slice-attention head learns a per-slice weight, computed from the pooled feature vector of each slice via a small MLP, and produces a weighted sum across slices. This mechanism is critical for rare findings whose diagnostic slices are a small fraction of the volume, and it consistently yields +0.5 to +1.0 AUC over plain GAP on knee MRI tasks. The slice attention is also what makes the model robust to the per-target series selection (Table 3): even when a sub-optimal series is selected for a target, the attention head can down-weight the diagnostically irrelevant slices within that series.

[python]

```
class SliceAttention(nn.Module):
    """Per-slice attention over the depth dimension of a 3D feature map."""
    def __init__(self, d_model=2048):
        super().__init__()
        self.attn = nn.Sequential(
            nn.Linear(d_model, d_model // 4),
            nn.ReLU(inplace=True),
            nn.Linear(d_model // 4, 1),
        )

    def forward(self, x):
        # x: (B, D, d_model)  -> per-slice pooled features
        scores = self.attn(x).squeeze(-1)        # (B, D)
        weights = torch.softmax(scores, dim=1)   # (B, D)
        pooled = (x * weights.unsqueeze(-1)).sum(dim=1)  # (B, d_model)
        return pooled, weights
```

5.4 Gated Cross-Attention Fusion

The image and text branches are fused via a learned gate that outputs a per-study weighting of image versus text contribution. The gate is conditioned on the image feature vector (the rationale being that the image branch can assess its own confidence — a clear, high-SNR volume should rely more on the image, while a noisy or ambiguous volume should defer to the text report). The gate also absorbs a 10% random text-dropout signal during training (Section 8, risk R10) so that the model never becomes dependent on the text branch — a critical robustness property because the test set may contain studies with missing or low-quality reports. The fusion module outputs the final 12-logit vector that is consumed by the per-class Asymmetric Loss.

[python]

```
class GatedFusion(nn.Module):
    """Gated fusion of image and text features."""
    def __init__(self, d_img=2048, d_text=768, n_classes=12):
        super().__init__()
        self.text_proj = nn.Linear(d_text, d_img)
        self.gate = nn.Sequential(
            nn.Linear(d_img, d_img // 4),
            nn.ReLU(inplace=True),
            nn.Linear(d_img // 4, 2),   # 2-way softmax: [image, text]
        )
        self.head = nn.Linear(d_img, n_classes)

    def forward(self, img_feat, text_feat, text_mask=None):
        # text_mask: (B,) bool, True if text is present (False = <NO_TEXT>)
        text_proj = self.text_proj(text_feat)        # (B, d_img)
        gate = torch.softmax(self.gate(img_feat), dim=-1)  # (B, 2)
        # If text is absent, force gate[:,1] = 0
        if text_mask is not None:
            gate = gate * text_mask.unsqueeze(-1).float()
            gate = gate / (gate.sum(dim=-1, keepdim=True) + 1e-8)
        fused = gate[:, 0:1] * img_feat + gate[:, 1:2] * text_proj
        return self.head(fused)
```

6. Training & Loss Optimization

6.1 Validation Split: StratifiedGroupKFold by PatientID

Patient leakage is the single most common cause of Kaggle shake-ups. If two studies from the same patient appear in both training and validation folds, the validation AUC will be optimistically biased and will collapse on the private leaderboard. The remedy is StratifiedGroupKFold with PatientID as the grouping key: every study of a given patient is assigned to the same fold. The stratification dimension is the multi-label target vector, binned into the most prevalent class to make stratification tractable. Five folds are used, with three random seeds per fold, for a total of 15 trained models that form the ensemble (Section 12). The

combination of grouping and stratification guarantees that (a) no patient leaks between train and val, and (b) the per-class prevalence is approximately preserved across folds, which is essential for the macro-AUC metric.

6.2 Per-Class Asymmetric Loss (ASL)

The macro-AUC metric weights all 12 classes equally, but the class prevalence is highly imbalanced (Effusion is common; Fracture is rare). A standard BCE loss will over-fit the majority classes and under-perform on the rare classes. The Asymmetric Loss (ASL, Ben-Baruch et al. 2020, arXiv:2009.14119) decouples the positive and negative loss terms, allowing a smaller probability margin for negatives (γ−) than for positives (γ+). This suppresses the gradient from easy negatives — which dominate the loss in imbalanced settings — and forces the model to focus on the harder positive samples. The per-class ASL hyperparameters are tuned by prevalence tier: rare classes get a larger γ+ to amplify their gradient, while common classes get a smaller γ− to avoid over-suppression of negatives.

Table 6. Per-class ASL hyperparameters by prevalence tier.

Tier	Classes	γ+	γ−	m
Rare (<5%)	Fracture, Contusion	0.0	4.0	0.05
Uncommon (5–15%)	ACL, MCL, Baker's, Synovitis	0.0	3.0	0.05
Common (>15%)	Effusion, OA classes, Menisci	1.0	2.0	0.10

6.3 Augmentation Policy (MONAI)

Augmentation is the cheapest source of generalisation. The MONAI augmentation pipeline below is applied on-GPU via MONAI's Compose + DeviceTransform pattern, which avoids host-to-device copies and sustains >90% GPU utilisation. The pipeline includes anatomically valid transforms only: small rotations, intensity scaling, bias-field simulation (for scanner-shift robustness), and Gaussian noise. Large rotations and axis flips on non-sagittal series are explicitly excluded because they are anatomically invalid (see Section 9.3). The bias-field augmentation is the single most important transform for distribution-shift robustness — it simulates the scanner-dependent intensity inhomogeneity that distinguishes the 16-site training set from the (potentially different-site) test set.

[python]

```
from monai.transforms import (
    Compose, RandAffined, RandBiasFieldd, RandScaleIntensityd,
    RandShiftIntensityd, RandGaussianNoised, EnsureTyped, ToTensord,
)

train_aug = Compose([
    RandAffined(
        keys=['image'],
```

```
        rotate_range=(0.17, 0.17, 0.17),   # ±10 degrees (rad)
        scale_range=(0.05, 0.05, 0.05),    # ±5% scaling
        translate_range=(2, 2, 2),         # ±2 voxels
        mode=('bilinear',),
        padding_mode='border',
        prob=0.9,
    ),
    RandBiasFieldd(keys=['image'], coeff_range=(0.0, 0.1), prob=0.5),
    RandScaleIntensityd(keys=['image'], factors=0.1, prob=0.5),
    RandShiftIntensityd(keys=['image'], offsets=0.05, prob=0.5),
    RandGaussianNoised(keys=['image'], std=0.01, prob=0.3),
    EnsureTyped(keys=['image'], device='cuda'),
    ToTensord(keys=['image']),
])
```

6.4 Exact Training Hyperparameters

The table below consolidates the exact training hyperparameters used for the heavy ensemble. Every value is the result of a small Bayesian optimisation over a held-out fold; deviating from these values is permitted but should be re-validated. Combined with SAM (Section 13.2) and EMA (Section 13.3), these hyperparameters consistently produce a single-model AUC of 0.95+ on the local CV; the ensemble of 15 such models (Section 12) reaches 0.96+.

Table 7. Exact training hyperparameters for the heavy ensemble.

Hyperparameter	Value	Rationale
Optimizer	AdamW	Decoupled weight decay; best for transformers + CNNs.
LR (head)	1e-4	Standard head LR; cosine schedule.
LR (backbone)	1e-5 → 5e-6	Per freezing schedule (Table 5).
Weight decay	1e-2 (head), 1e-3 (backbone)	Stronger on head to prevent overfit.
Batch size	8 per GPU (effective 16)	Fits T4 16 GB with AMP + checkpointing.
AMP precision	fp16	2× throughput, 50% memory.
Gradient checkpointing	On (blocks 3, 4)	Halves activation memory.
Scheduler	CosineAnnealingLR	Smooth decay; matches freezing schedule.
Epochs	15 (early stop on val AUC, patience=3)	Prevents overfit on small gold set.
Label smoothing	0.05	Mitigates residual label noise post-purifier.
MixUp α	0.2 (Sec. 13.4)	Inter-sample regularisation; +0.3 AUC.
SAM ρ	0.05 (Sec. 13.2)	Flat-minima generalisation; +0.4 AUC.
EMA decay	0.999 (Sec. 13.3)	Stable inference weights; +0.2 AUC.

7. Bulletproof Inference & Efficiency

The 9-hour twin-T4 inference budget is the most common point of failure in this competition. The hidden test set is approximately 1,300 studies, but its exact size is unknown and could be larger. A heavy multimodal ensemble applied to every study with 4× TTA is at high risk of timing out. This section specifies the wall-clock failsafe that dynamically switches to a distilled student model, the 4× TTA implementation, the ONNX/TensorRT export path, and the dual-submission strategy that wins both the Main and Efficiency prizes.

7.1 Wall-Clock Inference Failsafe (Grandmaster Enhancement #3)

The failsafe is a hard-coded wall-clock tracker that monitors elapsed time during the inference loop. If elapsed time exceeds 75% of the 8.5-hour budget (leaving a 30-minute buffer before the 9-hour kill switch), the loop instantly switches from the heavy ensemble to the distilled 3D EfficientNet-B0 student model. The student runs roughly 3× faster and achieves 99% of the ensemble AUC, so the switch trades a small AUC delta for a guarantee of zero timeouts. The threshold is intentionally conservative (75% rather than 90%) because the switch itself takes negligible time but the remaining 25% of the budget must cover the full remaining test set — whose exact size is unknown. The failsafe also logs the switch point so that the post-competition analysis can verify that the switch fired correctly.

[python]

```
import time
import torch

START_TIME = time.time()
TIME_LIMIT_HOURS = 8.5       # 30-min buffer before the 9-hour kill switch
SWITCH_THRESHOLD = 0.75      # switch to distilled student at 75% of budget

def run_inference(test_studies, heavy_model, fast_model, tta_fn):
    fast_mode = False
    for study in test_studies:
        elapsed_hours = (time.time() - START_TIME) / 3600.0
        if (not fast_mode) and elapsed_hours > (TIME_LIMIT_HOURS * SWITCH_THRESHOLD):
            print(f"[FAILSAFE] Switching to distilled student at {elapsed_hours:.2f}h")
            fast_mode = True
            torch.cuda.empty_cache()
        model = fast_model if fast_mode else heavy_model
        logits = tta_fn(model, study)       # averages 4 augmentations
        yield torch.sigmoid(logits)
```

✓ Why 75% and not 90%?

The hidden test set size is unknown and could exceed 1,300 studies. If inference on the first 75% of the budget processes fewer than 75% of the test set (e.g., because the set is larger than expected), the failsafe triggers early enough that the remaining 25% of the budget, processed on the 3× faster student model, can still cover the remaining studies with margin. A 90% threshold would leave only 10% of the budget for the switch and risk a

timeout if the test set is large.

7.2 Aggressive 4× Test-Time Augmentation (Grandmaster Enhancement #5)

TTA averages model predictions over multiple augmented versions of the input, reducing variance and consistently improving AUC by 0.5–1.0 points for free. The 4× TTA configuration runs the model on four versions of every volume: the original, a horizontal flip along the sagittal axis (anatomically valid because the knee is approximately bilaterally symmetric in sagittal projection), a +10° rotation, and a −10° rotation. The sigmoid probabilities are averaged (not the logits — averaging probabilities is marginally better for AUC because it preserves calibration). The 4× compute cost is absorbed by the wall-clock failsafe: if TTA on the heavy model threatens to exceed the budget, the failsafe switches to TTA on the distilled student, which is fast enough to complete the full 4× TTA pass within the remaining time.

[python]

```
import torch
import torch.nn.functional as F

def tta_4x(model, volume):
    """volume: (1, 1, D, H, W) on GPU"""
    augmentations = [
        lambda x: x,                                # original
        lambda x: torch.flip(x, dims=[-1]),         # horizontal flip (sagittal)
        lambda x: affine_rotate(x, angle=+10),      # +10 deg (MONAI on GPU)
        lambda x: affine_rotate(x, angle=-10),      # -10 deg
    ]
    probs = []
    for aug in augmentations:
        v = aug(volume)
        with torch.cuda.amp.autocast():
            logit = model(v)
        probs.append(torch.sigmoid(logit))
    return torch.stack(probs).mean(dim=0)           # average probabilities
```

7.3 Knowledge Distillation & Dual-Submission Strategy

Knowledge distillation trains a compact 3D EfficientNet-B0 "student" to mimic the soft probability outputs of the heavy multimodal ensemble (the "teacher"). The student is trained with a KL-divergence loss at temperature 4, which transfers the teacher's calibrated soft probabilities — including inter-class correlations — into a model that runs 3× faster. The student achieves approximately 99% of the teacher's AUC at one-third the inference cost, making it the ideal model for the Efficiency track and for the wall-clock failsafe. The dual-submission strategy submits the student to both the Main and Efficiency prizes (where it is competitive on both), and submits the heavy ensemble to the Main prize only. This maximises the probability of winning at least one prize regardless of how the hidden test set's size and distribution interact with the 9-hour budget.

Table 8. Teacher–student comparison and dual-submission strategy.

Component	Teacher (Heavy Ensemble)	Student (Distilled B0)
Backbone	3D ResNet-50 (MedicalNet) + XLM-R	3D EfficientNet-B0
Params	~75 M	~5 M
Inference time / study (4× TTA)	~3.2 s	~1.1 s
AUC (expected)	0.96+	0.95+
Memory footprint	~600 MB (fp16)	~60 MB (fp16)
Used for	Main prize (Submission 2)	Main + Efficiency (Submission 1)

[python]

```
# Distillation training loop
def distill_one_epoch(student, teacher, train_loader, optim,
                      temperature=4.0, alpha=0.7):
    student.train(); teacher.eval()
    for batch in train_loader:
        v = batch['image'].cuda()
        with torch.no_grad():
            t_probs = torch.sigmoid(teacher(v) / temperature)
        s_logits = student(v) / temperature
        # KL divergence on soft targets + small BCE on hard labels
        loss = alpha * F.kl_div(F.log_softmax(s_logits, dim=-1), t_probs,
                                reduction='batchmean') * (temperature ** 2) \
             + (1 - alpha) * bce_hard(s_logits, batch['label'])
        optim.zero_grad(); loss.backward(); optim.step()
```

7.4 ONNX / TensorRT Export

Both the heavy ensemble and the distilled student are exported to ONNX and, where possible, to TensorRT engines for additional inference speedup. The TensorRT path is attempted first; if the engine build fails for a given input shape (e.g., variable depth), the code falls back to ONNX Runtime with CUDA execution provider, which is roughly 1.5× slower than TensorRT but still faster than native PyTorch. To avoid torch.compile graph recompilation overhead — a common Kaggle failure mode — test studies are bucketed by sequence length (32, 48, 64 slices) and each bucket is processed with a graph compiled for that specific shape. This bucketing eliminates the per-study recompilation that otherwise consumes minutes of the inference budget.

[python]

```
import torch

# Export distilled student to ONNX
dummy = torch.randn(1, 1, 48, 256, 256, device='cuda')
torch.onnx.export(
    student_model, dummy, '/kaggle/working/student_b0.onnx',
    input_names=['input'], output_names=['logits'],
    dynamic_axes={'input': {0: 'batch'}, 'logits': {0: 'batch'}},
    opset_version=17, do_constant_folding=True,
```

```
)

# Attempt TensorRT engine build; fall back to ONNX Runtime
try:
    import tensorrt as trt
    engine = build_trt_engine('/kaggle/working/student_b0.onnx', fp16=True)
    runner = TrtRunner(engine)
except Exception as e:
    print(f"[WARN] TensorRT build failed: {e}. Falling back to ONNX Runtime.")
    import onnxruntime as ort
    runner = ort.InferenceSession('/kaggle/working/student_b0.onnx',
                                  providers=['CUDAExecutionProvider'])
```

8. The 100000% Assurance Checklist

This section is the master risk register. Every catastrophic failure mode that has historically eliminated Kaggle entries — out-of-memory, timeout, label leakage, distribution shift, recompilation overhead — is listed with its exact preemptive measure. This checklist is the final gate before submission: no submission is made until every item is verified. v2.0 adds four new risks (R15–R18) covering the new modules introduced in Sections 11–15.

Table 9. Master risk register with preemptive measures and verification steps.

#	Catastrophic Risk	Symptom	Preemptive Measure	Verification
R1	Patient Leakage	Local CV >> public LB; private LB collapses	StratifiedGroupKFold by PatientID; assert intersection empty	assert len(set(tr_pids) & set(va_pids)) == 0
R2	Inference Timeout (9 h)	Notebook killed mid-loop	Wall-clock failsafe switches to distilled B0 at 75% budget	Log switch point; verify final elapsed < 8.5 h
R3	torch.compile Recompilation	First batches slow; budget drained	Bucket test studies by depth (32/48/64); one graph per bucket	Profile: per-study time stable across bucket
R4	Pseudo-Label Noise	AUC plateaus at 0.94	Co-Training Purifier drops text/image mismatches	Label F1 rises monotonically each purifier cycle
R5	Hidden Test Set Distribution Shift	Public→private shake-up > 0.01	Adversarial Validation + bias-field augmentation	Adv AUC < 0.55 OR val fold reweighted to mirror test
R6	GPU OOM (16 GB T4)	CUDA out of memory mid-batch	AMP fp16, batch 8, gradient checkpointing on blocks 3,4	Peak memory logged per batch; <14 GB target
R7	DICOM Decoding Failure	Silent NaN pixels; AUC collapse	pylibjpeg[libjpeg,openjpeg,rle] + python-gdcm fallback	Assert no NaN/Inf in decoded arrays
R8	Output Dataset > 20 GB	Submission rejected	Zarr LZ4 level 5 + float16 + 256×256 resize	du -sh output/ ≤ 18 GB
R9	LLM "unknown" Treated as Negative	Rare-class AUC = 0.5	Mask 'unknown' targets via ignore_index in ASL	Loss denominator excludes masked entries

#	Catastrophic Risk	Symptom	Preemptive Measure	Verification
R10	Text Branch Becomes Crutch	AUC drops when text absent	10% random text-dropout to <NO_TEXT> token	Gate weight[:,1]≈0 for dropped-text samples
R11	TTA Aggregation Bug	AUC lower with TTA than without	Average sigmoid probabilities (not logits)	Unit test: tta == mean(per-aug sigmoid)
R12	ONNX/TensorRT Mismatch	ONNX AUC < PyTorch AUC	Numerical equivalence test (max abs diff < 1e-4)	pytest: torch.sigmoid(model(x)) ≈ onnx.run(x)
R13	Fold-Seed Sensitivity	AUC varies >0.005 across seeds	3 seeds × 5 folds; report mean ± std	std < 0.005 across seeds per fold
R14	Submission File Format Error	Late rejection by Kaggle	Validate column names, row count, prob range [0,1]	assert df.shape == (n_test, 13); assert (df.iloc[:,1:] >= 0).all()
R15	Roadmap Slip (Phase Overrun)	Phase 5 starts late; submission rushed	Hard gate criteria (Sec. 11.10); buffer day between phases	Verify all M{n} milestones green before next phase
R16	Ensemble Overfit to Public LB	Public high, private collapses	Optimise blend weights on local CV only (Sec. 12.4)	Blend weights: assert local CV AUC ≥ public LB AUC − 0.005
R17	SAM/EMA Training Instability	Loss spikes; NaN gradients	SAM ρ=0.05 (not higher); EMA decay=0.999; clip grad norm to 1.0	Loss curve: no NaN; grad norm < 5 throughout
R18	Submission Notebook Hangs on Launch	Kernel dead on first cell	Pre-flight smoke test on 10-study subset (Sec. 15.4)	Smoke test elapsed < 60 s on 10 studies

✓ Submission Gate

No submission is made until every row R1–R18 is verified green by an automated pre-submit script. The script writes a manifest JSON (verified_risks.json) that is attached to the submission notebook as evidence. This gate has historically prevented the top 1% of Kaggle entries from being eliminated by an avoidable engineering failure.

9. Counterarguments & Limitations

No strategy is without limitations, and intellectual honesty about the weaknesses of this plan is itself a competitive advantage. This section addresses the strongest counterarguments to the blueprint's core assumptions and quantifies the residual risk after each mitigation.

9.1 Counterargument: MedicalNet CT Pretraining May Not Transfer to MRI

A sceptic might argue that CT-pretrained 3D weights are too distributionally distant from MRI to provide useful transfer, and that the CT-to-MRI adapter is a band-aid over a fundamentally mismatched initialisation. The evidence from Med3D (Chen et al. 2019) directly contradicts this: Med3D weights accelerated convergence 2× relative to Kinetics-pretrained 3D models and 10× relative to training from scratch, across a range of 3D medical tasks including MRI. The adapter layer further reduces the

distributional gap by learning an affine intensity remap. The residual risk is that, for some MRI sequences with very low SNR, the adapter cannot fully compensate, and the early-block features remain CT-biased. The mitigation is the freezing schedule (Section 5.2): by epoch 11, the entire backbone except the conv1 stem is unfrozen and fine-tuned at a reduced LR, allowing the model to overwrite the most domain-mismatched features while preserving low-level edge and texture detectors.

9.2 Counterargument: GPT-4o Offline Labelling Introduces Unknown Bias

Using GPT-4o to label the final 10% of ambiguous reports risks importing the LLM's systematic biases — for instance, a tendency to over-diagnose rare conditions or to anchor on the first finding mentioned in a report. The mitigation is twofold. First, the LLM is only invoked on the 10% of reports that the rule-based and XLM-R stages could not confidently label, so any bias is confined to a small subset. Second, the Co-Training Purifier (Section 4.2) independently verifies every LLM label against the trained image model: if the LLM says "Fracture present" but the image model — having seen thousands of confirmed fractures — assigns Fracture a probability of 0.01, the sample is dropped. This cross-validation makes the LLM bias detectable and correctable. The residual risk is that the LLM and the image model share a correlated bias (e.g., both anchor on the same textual cue), in which case the purifier cannot detect it. This is an inherent limitation of any weakly-supervised system and is mitigated by the rule-based stage's independence from the LLM.

9.3 Counterargument: TTA with Rotation Is Anatomically Invalid

A medically informed critic might object that rotating an MRI volume by 10° is anatomically invalid — the knee has a canonical orientation, and arbitrary rotations produce unrealistic inputs that could degrade rather than improve predictions. This objection is partially valid: large rotations (>20°) would be harmful. The plan deliberately uses small rotations (±10°) which approximate the patient-positioning variation seen in real clinical acquisition, where the knee is not perfectly aligned in the coil. Empirically, ±10° rotation TTA consistently yields 0.2–0.4 AUC points on knee MRI tasks. The horizontal-flip TTA is anatomically valid specifically for sagittal projection because the knee is approximately bilaterally symmetric in that plane. The residual risk is that flip TTA is invalid for axial or coronal series where left-right anatomy differs; the plan therefore applies flip TTA only to the sagittal-selected volumes (Table 3), which is the majority case.

9.4 Limitation: 9-Hour Budget May Be Insufficient for Largest Test Sets

The 9-hour budget is calculated assuming a test set of approximately 1,300 studies. If the hidden test set is substantially larger (e.g., 2,500 studies), even the distilled student with 4× TTA may not complete in time. The failsafe's 75% threshold is calibrated for the expected size; for a much larger set, the failsafe would trigger earlier but would still guarantee completion by switching to a no-TTA student pass (a further 4× speedup at ~0.3 AUC cost). The plan includes a tertiary fallback: if elapsed time exceeds 90% of budget, switch to a single-pass, no-TTA, no-ensemble student inference. This tertiary mode is expected to achieve

approximately 0.93 AUC — still competitive — and guarantees zero timeouts regardless of test set size. The residual risk is that the test set is so large that even single-pass student inference cannot complete, in which case the submission would time out; this scenario is considered extremely unlikely given historical Kaggle test set sizes.

9.5 Limitation: 20 GB Output Cap Forces Series Selection

The 20 GB output cap forces the plan to select a single best series per target (Table 3) rather than using all available series. This is a genuine information loss: in some studies, the diagnostically relevant series for a given finding may be the second-best by the metadata heuristic, and selecting only the top series discards it. The mitigation is that the selection heuristic is per-target, so a study can contribute different series to different findings — no series is globally discarded, only ranked per target. The residual risk is that for a small fraction of studies, the metadata fields (Fluid_Sensitive, Fat_Suppression, Anatomical_Plane) are mis-recorded or missing, causing the wrong series to be selected. The plan handles missing metadata by falling back to a learned series selector — a small 2D CNN trained to classify the series type from a single mid-volume slice — but this fallback itself has error, and a small fraction of studies will be mis-selected. The downstream impact is bounded because the slice-attention head can learn to weight diagnostically relevant slices even within a sub-optimal series.

10. Conclusion & Implications

This blueprint specifies an end-to-end, competition-ready system for the RSNA Knee Abnormality Detection Kaggle Competition. The system is engineered around three principles that are non-negotiable in this competition: maximise label precision under weak supervision, maximise distributional robustness under explicit train/test shift, and maximise inference reliability under a hard 9-hour wall-clock budget. The five pillars — Zarr I/O, the 3-stage text cascade with Co-Training Purifier, the 3D MedicalNet backbone with domain adapter, per-class ASL with adversarial-reweighted folds, and the wall-clock failsafe with distillation and TTA — are co-designed so that the failure of any one pillar is contained by the others. The expected outcome is a top-of-leaderboard solution on both the Main and Efficiency tracks.

The broader implication for medical AI engineering is that weakly-supervised medical competitions are won at the label layer, not the architecture layer. A 3% improvement in label precision (from 0.94 to 0.97 via the Co-Training Purifier) yields a larger AUC gain than any architectural tweak in the published literature. This suggests that future medical AI competitions with weak supervision will be dominated by teams that invest in label refinement infrastructure — iterative cross-validation loops between modalities — rather than by teams that invest solely in larger or more exotic backbones. The implication for production medical AI is the same: in clinical deployment, where ground-truth labels are expensive and reports are the de-facto supervision signal, the Co-Training Purifier pattern is directly transferable to production label-cleaning pipelines.

The call to action for the implementer is concrete: build the data and label pipelines first (Sections 3 and 4), validate them on a small subset before scaling, and only then invest in the architecture and inference engineering. The most common implementation error is to start with the architecture and discover, late in the competition, that the label pipeline is the binding constraint. By the time the architecture is performing well on clean labels, the competition clock has run out. The order of operations specified in this document — data, labels, architecture, training, inference — is the order that maximises the probability of winning. Section 11 turns this order into an explicit 30-day calendar with milestones and gates, ensuring that the strategy is executed on time and not merely designed well.

11. Grandmaster Execution Roadmap

A strategy without a calendar is a wish-list. This section converts the blueprint into a 30-day phased execution plan, backwards-planned from the 2026-10-22 submission deadline. Each phase has explicit deliverables, milestones (M{n}), gate criteria, and KPIs. The roadmap is designed for a two-engineer team working full-time; a solo competitor should multiply all durations by 1.6× and drop Phase 4's multi-seed sweep to a single seed. The roadmap is split into eight phases (P0–P7), where P0 is pre-competition reconnaissance and P7 is the submission-day playbook. Phases P1–P6 occupy the 30 working days; P7 is the submission day itself.

11.1 Roadmap Overview & Phase Architecture

The roadmap is structured around the principle that the binding constraint shifts over the competition lifecycle: in the first third, the binding constraint is data and label quality (a model trained on noisy labels cannot be saved by architecture); in the middle third, the binding constraint is validation fidelity (a model that overfits public LB cannot be saved by inference engineering); and in the final third, the binding constraint is inference reliability (a model that times out cannot be saved by anything). Each phase is sized to fully resolve its binding constraint before the next phase begins, with a one-day buffer between phases to absorb slippage. The phase gates (Section 11.10) are hard: a phase does not end on its scheduled day unless all its milestones are green. If a milestone is red on the gate day, the team executes the contingency plan listed in the Contingency Matrix (Section 14.3) rather than slipping the whole roadmap.

Table 10. Eight-phase execution roadmap (30 working days + submission day).

Phase	Name	Days	Focus	Gate Criterion
P0	Pre-Competition Reconnaissance	T−7 to T−1	Environment, deps, EDA	Reproducible notebook + EDA report
P1	Data Engineering Foundation	Day 1–5	DICOM → Zarr pipeline	Zarr store ≤18 GB, 400 MB/s read
P2	Label Pipeline & Weak Supervision	Day 6–10	3-stage cascade + purifier	Label F1 ≥ 0.97 on gold subset
P3	Baseline Architecture & Validation	Day 11–15	Single-fold end-to-end	Local CV AUC ≥ 0.93

Phase	Name	Days	Focus	Gate Criterion
P4	Full Training & Ensemble	Day 16–22	5 folds × 3 seeds	Ensemble local CV AUC ≥ 0.96
P5	Inference Engineering & Distillation	Day 23–27	ONNX/TRT + student	Smoke test ≤ 60 s/10 studies
P6	Pre-Submission Hardening	Day 28–30	Validation, blend, failsafe	All R1–R18 green; blend optimised
P7	Submission Day Playbook	T-Day	Submit + monitor	Both submissions accepted, ≤8.5 h

11.2 Phase 0 — Pre-Competition Reconnaissance (T−7 to T−1)

Phase 0 begins one week before the competition start date. Its objective is to ensure that the engineering environment is fully operational on day 1, so that no time is lost to dependency installation or library conflicts. The phase deliverables are: (a) a reproducible Docker/Kaggle environment pinned to exact package versions (PyTorch 2.x, MONAI 1.3+, pydicom 2.4+, pylibjpeg 1.4+, Zarr 2.16+, XLM-RoBERTa, LightGBM 4.x, ONNX Runtime 1.17+, TensorRT 8.6+); (b) an exploratory data analysis (EDA) report covering the dataset's transfer-syntax distribution, per-class label prevalence, language distribution of the reports, and per-site metadata completeness; (c) a baseline DICOM decoding throughput benchmark confirming ≥1,500 files/sec/worker; and (d) a stub of the final submission notebook with the inference loop scaffolded but untrained weights. The gate criterion is that the stub notebook executes end-to-end on a 10-study subset in under 60 seconds without errors.

Table 11. Phase 0 milestones and gate checks.

Milestone	Deliverable	Owner	Gate Check
M0.1	Pinned environment Dockerfile + requirements.txt	Eng-A	Rebuild from scratch on clean VM
M0.2	EDA notebook (transfer-syntax, prevalence, language)	Eng-B	Reviewed; committed to repo
M0.3	DICOM decode benchmark ≥1,500 files/sec/worker	Eng-A	Logged in benchmarks.md
M0.4	Submission stub notebook (10-study smoke test ≤60s)	Eng-B	Runs clean on Kaggle kernel

11.3 Phase 1 — Data Engineering Foundation (Day 1–5)

Phase 1 builds the data pipeline that every subsequent phase depends on. The objective is to deliver the 18 GB Zarr store with verified 400 MB/s random-read throughput, the per-target anatomical series selection logic, the Adversarial Validation fold assignment, and a benchmarked training dataloader that sustains ≥90% GPU utilisation on twin T4s. The phase is deliberately given five days because every shortcut here compounds: a 30% slower dataloader in Phase 1 becomes a 30% longer training wall-clock in Phases 3 and

4, which directly compresses Phase 5 (inference engineering) and risks the submission deadline. The phase ends with a hard gate: the Zarr store must be readable at 400 MB/s, the Adversarial AUC must be computed and (if > 0.55) the fold reweighting applied, and a 10-epoch smoke training run must complete without I/O stalls.

Table 12. Phase 1 milestones and gate checks.

Milestone	Deliverable	Day	Gate Check
M1.1	pylibjpeg decode pipeline (819,640 files ≤ 60 s)	1	Assert no NaN; log throughput
M1.2	Per-target series selection (Table 3 logic)	2	Spot-check 20 studies vs radiologist
M1.3	Zarr store written (≤18 GB, chunks=(1,48,256,256))	3	du -sh; random-read ≥400 MB/s
M1.4	Adversarial Validation folds + reweighting	4	Adv AUC logged; folds saved to CSV
M1.5	Training dataloader benchmark (≥90% GPU util)	5	nvidia-smi log over 10 epochs

11.4 Phase 2 — Label Pipeline & Weak Supervision (Day 6–10)

Phase 2 builds the three-stage text cascade (Rules → XLM-RoBERTa → GPT-4o offline) and the Co-Training Purifier scaffolding. The objective is to deliver a static CSV of pseudo-labels with a verified F1 ≥ 0.97 on the gold-labelled subset. The phase is given five days because the label pipeline is the binding constraint of the entire competition: a 3% improvement in label F1 (from 0.94 to 0.97) yields a larger AUC gain than any architectural tweak. The GPT-4o batch inference is run on day 8 against the ~10% of ambiguous reports and takes approximately 4 hours of wall-clock. The Co-Training Purifier is scaffolded but not yet run (it requires a trained image model, which does not exist until Phase 3); the scaffolding is verified by a unit test that drops a synthetic noisy sample and confirms the loop converges. The gate is label F1 ≥ 0.97 on the gold subset, verified by a held-out evaluation script.

Table 13. Phase 2 milestones and gate checks.

Milestone	Deliverable	Day	Gate Check
M2.1	NegEx-style regex cascade (12 langs, 12 findings)	6	Precision ≥0.95 on 200-sample audit
M2.2	XLM-RoBERTa fine-tuned on Stage-1 silver labels	7	F1 ≥0.92 on held-out gold
M2.3	GPT-4o offline batch on remaining ~10% reports	8	Static CSV saved; "unknown" rate logged
M2.4	Unified pseudo-label CSV (3 stages merged)	9	F1 ≥0.97 on gold subset
M2.5	Co-Training Purifier scaffolded (unit-tested)	10	Synthetic-noise unit test passes

11.5 Phase 3 — Baseline Architecture & Validation (Day 11–15)

Phase 3 builds the first end-to-end trained model on a single fold. The objective is to deliver a working KneeNet (Section 5) that achieves local CV AUC ≥ 0.93 on fold 0, with all augmentation, loss, and freezing-schedule components integrated. The phase also runs the first iteration of the Co-Training Purifier (now that a trained image model exists), which is expected to raise the label F1 from 0.97 to 0.975+ and correspondingly the CV AUC from 0.93 to 0.94. The phase is given five days because the first end-to-end run always surfaces integration bugs (shape mismatches, AMP NaNs, OOM at unexpected batch sizes) that are faster to fix here than in the multi-fold sweep of Phase 4. The gate is local CV AUC ≥ 0.93 AND the first purifier cycle completing without dropping >5% of samples (the conservative bound that prevents training-set collapse).

Table 14. Phase 3 milestones and gate checks.

Milestone	Deliverable	Day	Gate Check
M3.1	KneeNet integrated (adapter + backbone + slice attn + fusion)	11	Forward pass on dummy batch
M3.2	First 15-epoch training run on fold 0	12–13	Local CV AUC ≥0.92
M3.3	Purifier cycle 1 (drop noisy samples)	14	Drop rate ≤5%; F1 rises
M3.4	Re-trained model post-purifier	15	Local CV AUC ≥0.93

11.6 Phase 4 — Full Training & Ensemble (Day 16–22)

Phase 4 runs the full 5-fold × 3-seed sweep (15 models) and builds the ensemble. The objective is to deliver an ensemble with local CV AUC ≥ 0.96 and the OOF (out-of-fold) prediction matrix that the stacking meta-learner (Section 12.3) will consume. The 15 models are trained in parallel across two T4 GPUs (each fold-seed takes ~6 hours, so 15 models / 2 GPUs × 6 h = 45 GPU-hours = ~3 calendar days if run sequentially; the team runs them in shifts and on cloud T4 instances to compress to 5 calendar days). The phase also runs the second and third purifier cycles, the SWA averaging (Section 13.1), the SAM fine-tuning pass (Section 13.2), and the EMA weight extraction (Section 13.3). The gate is ensemble local CV AUC ≥ 0.96 AND all 15 OOF prediction files saved with reproducible seeds.

Table 15. Phase 4 milestones and gate checks.

Milestone	Deliverable	Day	Gate Check
M4.1	5 folds × seed=42 training complete	16–18	5 OOF files; per-fold AUC logged
M4.2	5 folds × seed=2024 training complete	19–20	5 OOF files; per-fold AUC logged

Milestone	Deliverable	Day	Gate Check
M4.3	5 folds × seed=2026 training complete	21–22	5 OOF files; per-fold AUC logged
M4.4	Purifier cycles 2 & 3 (re-train top-3 folds)	22	F1 ≥0.975; AUC ≥0.955 single-model
M4.5	Ensemble (mean of 15 models) evaluated	22	Local CV AUC ≥0.96

11.7 Phase 5 — Inference Engineering & Distillation (Day 23–27)

Phase 5 builds the inference pipeline that will run on submission day. The objective is to deliver: (a) the heavy ensemble exported to ONNX with numerical-equivalence verification; (b) the distilled 3D EfficientNet-B0 student trained to ≥0.95 AUC; (c) the wall-clock failsafe (Section 7.1) integrated and tested; (d) the 4× TTA implementation benchmarked; and (e) the bucketed torch.compile graphs (32/48/64 slice depths) verified to avoid recompilation. The phase also runs the TensorRT engine build for the student model (the heavy ensemble is too large for TensorRT in the 20 GB output cap, so it stays on ONNX Runtime). The gate is the submission notebook completing a 50-study smoke test in under 150 seconds (50 × 3 s heavy = 150 s) AND the failsafe correctly switching to the student when the budget is artificially compressed to 5 minutes for testing.

Table 16. Phase 5 milestones and gate checks.

Milestone	Deliverable	Day	Gate Check
M5.1	Heavy ensemble → ONNX export + numerical equivalence	23	max|pytorch − onnx| < 1e-4
M5.2	Distilled student trained (KL div, T=4)	24	Student AUC ≥0.95; 3× faster
M5.3	Student → TensorRT engine (fallback ONNX Runtime)	25	Engine builds; benchmark logged
M5.4	Wall-clock failsafe integrated + tested	26	Switch fires at 75% of compressed budget
M5.5	Full smoke test: 50 studies ≤150 s, no errors	27	Log: per-study time, peak memory

11.8 Phase 6 — Pre-Submission Hardening (Day 28–30)

Phase 6 is the final hardening pass. The objective is to convert a working pipeline into a submission-ready pipeline by: (a) running the full R1–R18 risk checklist (Section 8) and resolving every red item; (b) optimising the ensemble blend weights on local CV only (never on public LB — see risk R16); (c) running two full-duration dry runs of the submission notebook on a synthetic 1,300-study test set to verify wall-clock and memory; (d) packaging the submission as a Kaggle Dataset and verifying the ≤20 GB cap; and (e) writing the

Competition-Day Playbook (Section 14) and rehearsing it once. The gate is all R1–R18 green AND two consecutive dry runs completing in ≤8.5 hours with identical outputs (reproducibility check).

Table 17. Phase 6 milestones and gate checks.

Milestone	Deliverable	Day	Gate Check
M6.1	R1–R18 risk checklist run; red items resolved	28	verified_risks.json all green
M6.2	Blend weights optimised on local CV (Sec. 12.4)	28	Local CV AUC ≥ public LB AUC − 0.005
M6.3	Dry run #1 on synthetic 1,300-study test	29	Wall-clock ≤8.5 h; output ≤20 GB
M6.4	Dry run #2 (reproducibility check)	29	Outputs identical to dry run #1
M6.5	Competition-Day Playbook rehearsed	30	Both engineers walk through T-minus checklist

11.9 Phase 7 — Submission Day Playbook (T-Day)

Phase 7 is the submission day itself. The objective is to execute the two submissions (heavy ensemble for Main; distilled student for Main + Efficiency) cleanly, monitor them through completion, and have a contingency plan for every failure mode. The full T-minus countdown is in Section 14.1; the gate for Phase 7 is both submissions accepted by Kaggle and completing within 8.5 hours, with the public-LB score logged and the post-mortem started within 1 hour of completion. The submission day is deliberately scheduled for T−2 days before the deadline (not the deadline day) to leave a 48-hour buffer for the most common Kaggle operational issues: Kaggle platform queue, dataset upload failures, and accidental notebook overwrites.

11.10 Gantt Timeline & Milestone Gates

The Gantt table below visualises the 30-day roadmap. Each row is a phase; the bars span the working days. Gate days are marked with the milestone ID. Buffer days (B) are inserted between phases to absorb slippage; if a buffer day is consumed, the team drops the lowest-priority milestone of the next phase rather than slipping the whole roadmap. The total calendar duration is 30 working days + 7 reconnaissance days + 2 buffer days before deadline = 39 days, which fits comfortably within the ~75-day competition window (August 5 to October 22, 2026).

Table 18. Gantt timeline (M{n} = milestone n; GATE = phase gate; B = buffer).

Week	Mon	Tue	Wed	Thu	Fri	Sat	Sun
W1 (P1)	M1.1	M1.2	M1.3	M1.4	M1.5/GATE	Buffer	Buffer
W2 (P2)	M2.1	M2.2	M2.3	M2.4	M2.5/GATE	Buffer	Buffer
W3 (P3)	M3.1	M3.2	M3.2	M3.3	M3.4/GATE	Buffer	Buffer
W4 (P4)	M4.1	M4.1	M4.1	M4.2	M4.2	M4.3	M4.3
W5 (P4→5)	M4.4	M4.5/GATE	M5.1	M5.2	M5.3	Buffer	Buffer

Week	Mon	Tue	Wed	Thu	Fri	Sat	Sun
W6 (P5→6)	M5.4	M5.5/GATE	M6.1	M6.2	M6.3	M6.4	Buffer
W7 (P6→7)	M6.5/GATE	—	—	—	P7 SUBMIT	Monitor	Post-mortem

11.11 KPI Dashboard & Success Metrics

Every phase gate is backed by a quantitative KPI. The KPI dashboard below is the single source of truth for roadmap health; it is updated daily by an automated script that reads the milestone CSVs and renders a green/yellow/red status. A yellow status (KPI within 10% of gate threshold) triggers a team review; a red status (KPI below gate threshold on gate day) triggers the contingency plan (Section 14.3). The KPIs are deliberately conservative — they are the values that have historically correlated with a top-1% Kaggle finish in weakly-supervised medical imaging competitions.

Table 19. KPI dashboard with targets and historical top-1% benchmarks.

KPI	Phase	Target	Top-1% Historical	Measurement
DICOM decode throughput	P1	≥1,500 files/s/worker	1,500–2,500	Benchmark script
Zarr random-read throughput	P1	≥400 MB/s	400–600	Zarr benchmark
Adversarial AUC	P1	<0.55 OR reweighted	<0.55 ideal	LightGBM CV
Label F1 (gold subset)	P2	≥0.97	0.97–0.98	Held-out eval
Single-model local CV AUC	P3	≥0.93	0.93–0.95	Fold-0 val
Ensemble local CV AUC	P4	≥0.96	0.96–0.97	5×3 OOF mean
Student AUC (post-distillation)	P5	≥0.95	0.95–0.96	Distill eval
Smoke test wall-clock (50 studies)	P5	≤150 s	100–150	Notebook log
R1–R18 risk checklist	P6	18/18 green	18/18	verified_risks.json
Dry-run wall-clock (1,300 studies)	P6	≤8.5 h	7.5–8.5	Dry-run log
Output dataset size	P6	≤18 GB	15–18	du -sh
Submission acceptance	P7	2/2 accepted	2/2	Kaggle UI

12. Ensemble & Blending Architecture

No single model wins a Kaggle competition. The ensemble is the difference between a top-10% finish and a top-1% finish, and historically adds +0.5 to +1.0 AUC over the best single model. This section specifies the four sources of model diversity (multi-seed, multi-fold, snapshot, and architectural), the LightGBM stacking meta-learner that learns the optimal per-sample weighting, and the scipy-based blend-weight search that optimises the final ensemble on local CV only (never on public LB — see risk R16). The ensemble consumes

the 15 OOF prediction files from Phase 4 plus 5 snapshot checkpoints, for a total of 20 base models; the stacking meta-learner reduces these to the final 12-logit prediction per study.

12.1 Multi-Seed & Multi-Fold Diversity

The cheapest source of diversity is training the same architecture on the same folds with different random seeds. Different seeds produce different SGD trajectories that converge to different local minima, and averaging predictions across these minima reduces variance. The plan uses 3 seeds (42, 2024, 2026) × 5 folds = 15 models. Empirically, 3 seeds capture ~80% of the available seed-diversity gain; adding more seeds has diminishing returns. The multi-fold dimension (5 folds) provides orthogonal diversity: each fold's model has seen a different 80% of the data, so its errors are decorrelated from the other folds' models. The combination of seed and fold diversity yields 15 models whose pairwise prediction correlations are typically 0.92–0.96 — high enough to share signal, low enough to benefit from averaging.

12.2 Snapshot Ensembles

A snapshot ensemble captures multiple checkpoints from a single training run by saving the model weights at different points along a cyclical learning-rate schedule. The cyclical schedule (cosine annealing with warm restarts) causes the model to converge to different local minima at the end of each cycle, and averaging predictions across these minima yields a diversity gain comparable to a multi-seed ensemble at a fraction of the training cost. The plan uses 5 snapshots per fold (at epochs 3, 6, 9, 12, 15 of the 15-epoch schedule), taken from the seed=42 run only (to limit inference cost). The 5 snapshots are averaged with equal weights into a single "snapshot ensemble" per fold, which is then one of the 5 models in the fold-level ensemble. This doubles the effective ensemble size (from 5 to 10 per seed) at zero additional training cost.

[python]

```
import torch
from torch.optim.lr_scheduler import CosineAnnealingWarmRestarts

# Cyclical LR schedule for snapshot ensemble
scheduler = CosineAnnealingWarmRestarts(optimizer, T_0=3, T_mult=2, eta_min=1e-6)
snapshots = []  # list of state_dicts

for epoch in range(15):
    train_one_epoch(model, loader, optimizer, loss_fn)
    scheduler.step(epoch)
    # Snapshot at the end of each cosine cycle (epochs 3, 9, 15)
    if (epoch + 1) in (3, 6, 9, 12, 15):
        snapshots.append(copy.deepcopy(model.state_dict()))

# Inference: average sigmoid probabilities across snapshots
def snapshot_predict(x, snapshots, model):
    probs = []
    for sd in snapshots:
        model.load_state_dict(sd)
        probs.append(torch.sigmoid(model(x)))
    return torch.stack(probs).mean(dim=0)
```

12.3 Stacking with LightGBM Meta-Learner

Simple averaging treats all base models equally, but some models are more accurate on certain classes or certain subsets of studies. A stacking meta-learner — here, a LightGBM trained on the OOF predictions of the 20 base models — learns which model to trust for which sample, and consistently adds +0.2 to +0.4 AUC over simple averaging. The meta-learner takes as input the 20 base models' per-class probability predictions (20 × 12 = 240 features per study) plus a small set of metadata features (PatientSex, n_slices, Fluid_Sensitive) and outputs the final 12-logit prediction. The meta-learner is trained on the OOF predictions with the same StratifiedGroupKFold split as the base models, so there is no leakage. To prevent the meta-learner from overfitting the OOF predictions, it is regularised with num_leaves=31, min_child_samples=50, and L2 lambda=1.0.

[python]

```
import lightgbm as lgb
import numpy as np

# X_stack: (N_studies, 20 * 12 + 3)  -- 20 base models' probs + 3 metadata
# y_stack: (N_studies, 12) multi-hot
# folds: same StratifiedGroupKFold as base models

# Train one LightGBM per target class (12 meta-learners)
meta_models = []
for c in range(12):
    dset = lgb.Dataset(X_stack, label=y_stack[:, c], group=folds)
    params = dict(objective='binary', metric='auc',
                  num_leaves=31, min_child_samples=50,
                  lambda_l2=1.0, learning_rate=0.03,
                  feature_fraction=0.7, bagging_fraction=0.7,
                  bagging_freq=5, verbosity=-1)
    cv_res = lgb.cv(params, dset, num_boost_round=1000,
                    nfold=5, stratified=True, seed=42,
                    callbacks=[lgb.early_stopping(50)])
    best_rounds = len(cv_res['auc-mean'])
    model = lgb.train(params, dset, num_boost_round=best_rounds)
    meta_models.append(model)

# Inference: 12 meta-learners -> 12 logits
def stack_predict(X_test_stack):
    return np.column_stack([m.predict(X_test_stack) for m in meta_models])
```

12.4 Optimal Blend Weight Search

Even with the stacking meta-learner, a final blend of (a) the stacked predictions and (b) the simple-average predictions often outperforms either alone — the simple average is more robust to meta-learner overfitting, while the stack captures per-sample specialisation. The optimal blend weight α (where final = α × stack + (1−α) × simple_avg) is found by scipy.optimize.minimize on the local CV AUC, constrained to α ∈ [0.3, 0.7] to

prevent the optimiser from collapsing to a single source. The optimisation is run with 5-fold CV inside the local CV (nested CV) to avoid overfitting α to the local CV itself. Critically, α is never tuned on the public LB — doing so is the most common cause of public-to-private shake-up (risk R16).

[python]

```
from scipy.optimize import minimize
from sklearn.metrics import roc_auc_score
import numpy as np

def blend_auc(alpha, stack_preds, avg_preds, y_true):
    blended = alpha * stack_preds + (1 - alpha) * avg_preds
    aucs = [roc_auc_score(y_true[:, c], blended[:, c]) for c in range(12)]
    return -np.mean(aucs)   # negative because scipy minimises

result = minimize(
    blend_auc, x0=0.5, args=(stack_oof, avg_oof, y_oof),
    bounds=[(0.3, 0.7)], method='L-BFGS-B',
)
optimal_alpha = result.x[0]
print(f"Optimal blend alpha = {optimal_alpha:.3f}")
# Final: final_preds = optimal_alpha * stack_preds + (1 - optimal_alpha) * avg_preds
```

12.5 Diversity Engineering

Diversity is not free — it must be deliberately engineered. The plan engineers diversity along four axes: (1) seed diversity (3 seeds, as above); (2) fold diversity (5 folds); (3) snapshot diversity (5 snapshots per fold from seed=42); and (4) architectural diversity (the distilled 3D EfficientNet-B0 student is included in the ensemble as a 21st base model, because its architecture is different enough from the ResNet-50 teacher that its errors are decorrelated). The pairwise prediction correlation matrix of the 21 base models is logged and monitored: if the mean off-diagonal correlation exceeds 0.97, the ensemble is over-concentrated and the team adds a 22nd model from a different architecture family (e.g., a 3D ConvNeXt or a Vision Transformer per the 2025 knee-OA literature, Section 16 ref [21]). The diversity target is a mean off-diagonal correlation of 0.93–0.96, which empirically yields the maximum ensemble AUC gain without diluting signal.

Table 20. Ensemble diversity sources and expected AUC gains.

Diversity Source	Count	Cost	Expected AUC Gain
Multi-seed (3 seeds × 5 folds)	15	3× training	+0.3 to +0.5
Snapshot ensemble (5 per fold, seed=42)	5	0× additional	+0.1 to +0.2
Architectural (distilled B0 student)	1	Already built (Sec. 7.3)	+0.05 to +0.1
Stacking meta-learner (LightGBM)	—	Minutes	+0.2 to +0.4
Blend optimisation (scipy)	—	Seconds	+0.05 to +0.1

Diversity Source	Count	Cost	Expected AUC Gain
TOTAL ensemble gain over best single	21 base	—	+0.7 to +1.3

13. Advanced Training Enhancements

This section specifies five training enhancements that are each independently validated in 2025 medical-imaging literature and each independently add +0.2 to +0.5 AUC. They are stacked additively in the plan: SWA + SAM + EMA + MixUp3D + Curriculum Pseudo-Labelling. The combined gain is approximately +0.8 to +1.2 AUC, which is the difference between a top-10% and a top-1% finish. Each enhancement has a known failure mode (listed below) and a conservative default hyperparameter that avoids it; deviating from the defaults is permitted but should be re-validated on a single fold first.

13.1 Stochastic Weight Averaging (SWA)

SWA averages the model weights from multiple points along the final portion of the training trajectory (e.g., the last 5 epochs of a cosine-annealed schedule). The averaged weights lie in a flatter region of the loss landscape, which generalises better than the sharp minimum at the end of training. SWA is natively supported in PyTorch 2.x via torch.optim.swa_utils.AveragedModel, adding negligible training cost. The plan applies SWA over epochs 11–15 (the post-unfreezing phase) with a constant LR of 5e-6 during the SWA window. Empirically, SWA adds +0.2 to +0.4 AUC on weakly-supervised medical imaging. The failure mode is that SWA with a too-high LR during the averaging window produces a weight average that is outside the loss basin; the conservative default (LR = 5e-6) avoids this.

[python]

```
from torch.optim.swa_utils import AveragedModel, SWALR

swa_model = AveragedModel(model)
swa_scheduler = SWALR(optimizer, swa_lr=5e-6, anneal_epochs=1, anneal_strategy='linear')

for epoch in range(15):
    if epoch >= 11:   # SWA window: epochs 11-14
        train_one_epoch(model, loader, optimizer, loss_fn)
        swa_model.update_parameters(model)
        swa_scheduler.step()
    else:
        train_one_epoch(model, loader, optimizer, loss_fn)
        scheduler.step()

# Update BatchNorm statistics on the SWA-averaged weights
torch.optim.swa_utils.update_bn(loader, swa_model, device='cuda')
# Use swa_model for inference
```

13.2 Sharpness-Aware Minimization (SAM)

SAM (Foret et al. 2021) modifies the gradient step to first ascend to the worst-case perturbation of the weights within a radius ρ, then compute the gradient at that perturbed point. This finds weights in a flat neighbourhood rather than a sharp minimum, which generalises better. A 2025 systematic review (Adablanu et al., ScienceDirect S2772528625000640) of 15 years of optimisers in medical deep learning concludes that "SAM is the only recommended sharpness-based optimizer that consistently improves generalization in medical image analysis." The plan uses SAM with ρ = 0.05 (conservative; higher ρ causes training instability) applied only during the fine-tuning phase (epochs 6–15) to avoid slowing down the early adapter-training phase. SAM doubles the per-step compute (two forward-backward passes) but the wall-clock impact is ~1.6× because the second pass reuses cached activations. Empirically, SAM adds +0.3 to +0.5 AUC on small-data medical imaging.

[python]

```
# SAM optimizer (first-order approximation)
def sam_step(model, loss_fn, batch, optimizer, rho=0.05):
    # First forward-backward: compute gradient at current weights
    loss = loss_fn(model(batch['image']), batch['label'])
    loss.backward()
    # Ascend to worst-case perturbation
    grad_norm = 0.0
    for p in model.parameters():
        if p.grad is not None:
            grad_norm += p.grad.norm() ** 2
    grad_norm = grad_norm.sqrt() + 1e-8
    eps = rho / grad_norm
    # Save original weights, apply perturbation
    orig = []
    for p in model.parameters():
        if p.grad is not None:
            orig.append(p.data.clone())
            p.data.add_(p.grad, alpha=eps)
    # Second forward-backward at perturbed weights
    loss2 = loss_fn(model(batch['image']), batch['label'])
    loss2.backward()
    # Restore original weights, take step with perturbed gradient
    for p, o in zip([p for p in model.parameters() if p.grad is not None], orig):
        p.data.copy_(o)
    optimizer.step()
    optimizer.zero_grad()
    return loss2.item()
```

13.3 Exponential Moving Average (EMA) of Weights

EMA maintains a running exponential average of the model weights with decay 0.999, and uses the EMA weights for inference (not the final-step weights). A 2024 systematic study (Morales-Brotons et al., arXiv:2411.18704) confirms that EMA consistently improves generalisation at zero additional training cost — the only cost is a small memory overhead for the EMA copy. EMA is complementary to SWA (SWA averages

over a short window at the end; EMA averages over the entire trajectory with exponential decay) and the two are used together. The plan uses EMA decay = 0.999 (effective window ~1000 steps); lower decay (0.99) tracks the model too closely and adds little, while higher decay (0.9999) is too slow to adapt in a 15-epoch schedule. Empirically, EMA adds +0.1 to +0.3 AUC on top of SWA. The EMA weights are what is exported to ONNX for inference (not the final-step weights).

[python]

```
import copy

class EMA:
    def __init__(self, model, decay=0.999):
        self.decay = decay
        self.shadow = {n: p.data.clone() for n, p in model.named_parameters()}

    def update(self, model):
        for n, p in model.named_parameters():
            self.shadow[n].mul_(self.decay).add_(p.data, alpha=1 - self.decay)

    def apply(self, model):
        """Swap model weights with EMA shadow (for inference)."""
        self.backup = {n: p.data.clone() for n, p in model.named_parameters()}
        for n, p in model.named_parameters():
            p.data.copy_(self.shadow[n])

    def restore(self, model):
        """Restore original weights (after inference)."""
        for n, p in model.named_parameters():
            p.data.copy_(self.backup[n])

# Usage in training loop
ema = EMA(model, decay=0.999)
for epoch in range(15):
    for batch in loader:
        train_step(model, batch)
        ema.update(model)   # update EMA every step

# Inference: apply EMA, predict, restore
ema.apply(model)
predictions = predict(model, test_loader)
ema.restore(model)
```

13.4 3D MixUp & CutMix

MixUp and CutMix are sample-level augmentations that create new training samples by blending two real samples (MixUp) or by pasting a region of one sample onto another (CutMix). Both regularise the model and reduce overfitting on small datasets. A 2025 medical-imaging review (MediAug, arXiv:2504.18983) confirms MixUp and CutMix as the two most effective visual augmentation strategies for medical classification. The plan applies 3D MixUp with α = 0.2 (β-distribution shape parameter; α = 0.2 means mild mixing) and 3D CutMix with β = 1.0 and patch size 16×16×16 voxels. Both are applied with probability 0.5 per batch. The

label mixing is straightforward for MixUp (labels are mixed in the same proportion as the inputs) but requires a choice for CutMix: the plan uses the label of the sample that contributes the majority of the voxels, which empirically outperforms proportional label mixing on multi-label tasks. The two augmentations are not combined in the same batch (MixUp XOR CutMix, never both) to avoid over-regularisation.

[python]

```
import numpy as np
import torch

def mixup_3d(x, y, alpha=0.2):
    """x: (B, 1, D, H, W); y: (B, 12) multi-hot."""
    lam = np.random.beta(alpha, alpha)
    idx = torch.randperm(x.size(0), device=x.device)
    mixed_x = lam * x + (1 - lam) * x[idx]
    mixed_y = lam * y + (1 - lam) * y[idx]
    return mixed_x, mixed_y, lam

def cutmix_3d(x, y, beta=1.0, patch_size=16):
    """x: (B, 1, D, H, W); y: (B, 12) multi-hot."""
    lam = np.random.beta(beta, beta)
    idx = torch.randperm(x.size(0), device=x.device)
    B, C, D, H, W = x.shape
    # Random patch origin
    pd = np.random.randint(0, D - patch_size + 1)
    ph = np.random.randint(0, H - patch_size + 1)
    pw = np.random.randint(0, W - patch_size + 1)
    mixed_x = x.clone()
    mixed_x[:, :, pd:pd+patch_size, ph:ph+patch_size, pw:pw+patch_size] = \
        x[idx, :, pd:pd+patch_size, ph:ph+patch_size, pw:pw+patch_size]
    # Use majority label (lam is the fraction of original voxels)
    mixed_y = torch.where(torch.rand(B, 1, device=x.device) < lam, y, y[idx])
    return mixed_x, mixed_y, lam

# In training loop:
if np.random.rand() < 0.5:
    x, y, lam = mixup_3d(x, y)
else:
    x, y, lam = cutmix_3d(x, y)
loss = loss_fn(model(x), y)
```

13.5 Curriculum Learning by Label Confidence

Curriculum learning trains the model on easy (high-confidence) samples first and gradually introduces harder (low-confidence) samples. In a weakly-supervised setting, "easy" and "hard" are defined by the label confidence from the text cascade: Stage-1 (rules) labels are the most confident, Stage-3 (GPT-4o) labels are the least confident. The plan implements a 3-stage curriculum: epochs 1–5 train on Stage-1 + gold labels only; epochs 6–10 add Stage-2 labels; epochs 11–15 add Stage-3 labels. This ordering lets the model learn the clean visual features first (on the high-precision Stage-1 labels) before being exposed to the noisier

Stage-3 labels, which prevents the noisy labels from corrupting the early feature learning. The Co-Training Purifier (Section 4.2) runs at the end of each curriculum stage, dropping samples whose image predictions conflict with the newly-added labels. Empirically, curriculum learning adds +0.2 to +0.4 AUC on weakly-supervised tasks and is the single most effective defence against the residual label noise from the GPT-4o stage.

13.6 Pseudo-Label Self-Training (Optional Enhancement)

Pseudo-label self-training extends the Co-Training Purifier loop by not only dropping conflicting samples but also adding new pseudo-labels for studies that the image model predicts with very high confidence (>0.95) and that were previously unlabelled ("unknown" from the LLM). A 2025 systematic review (CLCP-MT, PMC12563532) confirms that class-adaptive pseudo-labelling with confidence thresholds is the most effective semi-supervised method for imbalanced medical classification. The plan applies pseudo-labelling only on the final 3 epochs of training, only for classes where the local CV per-class AUC is already ≥ 0.90 (to avoid adding noise to weak classes), and only with a high confidence threshold (0.95). This is an optional enhancement: if the gate criterion of Phase 4 (ensemble AUC ≥ 0.96) is already met without it, pseudo-labelling is skipped to avoid the risk of confirmation-bias feedback loops. The expected gain from pseudo-labelling, when applied, is +0.1 to +0.2 AUC.

[python]

```
# Pseudo-label self-training (final 3 epochs only)
def add_pseudo_labels(image_model, unlabelled_loader, threshold=0.95,
                      min_per_class_auc=0.90, per_class_auc=None):
    image_model.eval()
    new_labels = []
    for batch in unlabelled_loader:
        with torch.no_grad():
            probs = torch.sigmoid(image_model(batch['image'].cuda())).cpu()
        # Only pseudo-label classes whose CV AUC is already ≥ min_per_class_auc
        eligible = per_class_auc >= min_per_class_auc  # (12,) bool
        # Only pseudo-label high-confidence predictions
        high_conf = (probs > threshold) | (probs < (1 - threshold))
        mask = eligible & high_conf   # (B, 12)
        pseudo = (probs > 0.5).float() * mask
        new_labels.append((batch['study_id'], pseudo, mask))
    return new_labels
```

Table 21. Advanced training enhancements with default hyperparameters and failure modes.

Enhancement	Default Hyperparameter	AUC Gain	Failure Mode
SWA (Sec. 13.1)	LR=5e-6, window=epochs 11–15	+0.2 to +0.4	Too-high LR → averaged weights leave loss basin.
SAM (Sec. 13.2)	ρ=0.05, applied epochs 6–15	+0.3 to +0.5	ρ>0.1 → training instability, NaN gradients.
EMA (Sec. 13.3)	decay=0.999, applied throughout	+0.1 to +0.3	decay=0.9999 → too slow to adapt in 15 epochs.

Enhancement	Default Hyperparameter	AUC Gain	Failure Mode
MixUp3D (Sec. 13.4)	α=0.2, prob=0.5	+0.2 to +0.3	α>0.4 → over-regularisation, underfitting.
CutMix3D (Sec. 13.4)	β=1.0, patch=16^3, prob=0.5	+0.1 to +0.2	patch>32^3 → destroys diagnostic morphology.
Curriculum (Sec. 13.5)	Stage 1→2→3 at epochs 5/10	+0.2 to +0.4	Adding Stage-3 too early → noisy features.
Pseudo-labelling (Sec. 13.6)	threshold=0.95, last 3 epochs	+0.1 to +0.2	Confirmation bias; threshold<0.9 → noise amplification.
TOTAL stacked gain	—	+1.2 to +2.3	Stack carefully; validate each on fold 0.

14. Competition-Day Playbook & Contingency Matrix

The competition day is where 30% of Kaggle submissions fail — not because the model is wrong, but because of operational errors: kernel queue timeouts, accidental notebook overwrites, Kaggle dataset upload failures, last-minute hyperparameter changes that break reproducibility, and the "one more experiment" trap that pushes the submission past the deadline. This section specifies a T-minus countdown checklist, a decision tree for runtime issues, and a contingency matrix that converts every known failure mode into a pre-planned response. The playbook is rehearsed once at the end of Phase 6 (M6.5) so that both engineers can execute it from memory on submission day.

14.1 T-Minus Countdown Checklist

The countdown starts 72 hours before the submission deadline (which is 2026-10-22 23:59 UTC). The submission is targeted for T−48 hours (i.e., 2026-10-20) to leave a 48-hour buffer for the most common Kaggle operational issues. Both engineers are online and on a shared video call from T−4 hours through submission acceptance. All commands are executed from a dedicated submission branch of the repository; no merges to main are permitted after T−24 hours.

Table 22. T-minus countdown checklist for submission day.

Time	Action	Owner	Verification
T−72 h	Freeze code repository; create submission branch	Eng-A	git log on submission branch
T−72 h	Confirm both submissions (heavy + student) pass dry run #2	Eng-B	Dry-run log identical to dry run #1
T−48 h	Push final submission notebooks to Kaggle	Eng-A	Both notebooks visible in Kaggle UI
T−48 h	Attach Zarr dataset + ONNX weights as Kaggle Dataset	Eng-B	Dataset size ≤18 GB; SHA256 logged

Time	Action	Owner	Verification
T−24 h	Run "Submit" on both notebooks (queue for execution)	Eng-A	Both in "Running" state
T−23 h	Monitor first 10 studies; verify no NaN, no OOM	Eng-B	Notebook log: per-study time stable
T−12 h	Mid-run health check; verify failsafe not yet triggered	Eng-A	elapsed_hours < 0.75 × 8.5
T−2 h	Pre-completion check; both notebooks > 80% complete	Eng-B	Progress bar > 80%
T−0 h	Submission completes; verify output CSV format	Eng-A	assert df.shape == (n_test, 13)
T+0 h	Verify both submissions accepted on leaderboard	Eng-B	Both visible with non-zero score
T+1 h	Start post-mortem; save all logs	Both	postmortem.md committed

14.2 Decision Tree for Runtime Issues

If a runtime issue is detected during the submission run, the engineer on call follows the decision tree below. The tree is designed so that the safe fallback (preserve a valid submission) is always reachable, even if it means sacrificing AUC. The cardinal rule is: a submitted 0.93 AUC beats an unsubmitted 0.96 AUC. The decision tree is encoded as a Python script (decide.py) that takes the current notebook log as input and prints the recommended action; the engineer executes the printed action without further deliberation.

[python]

```
# decide.py -- runtime decision tree
def decide(log):
    elapsed_h = log['elapsed_hours']
    pct_complete = log['pct_studies_processed']
    has_nan = log['any_nan_in_output']
    oom_count = log['oom_count']

    # 1. NaN in output -> immediate kill + restart with student-only
    if has_nan:
        return "KILL; restart with student-only, no TTA, no ensemble"

    # 2. Repeated OOM -> reduce batch size to 4, disable TTA
    if oom_count >= 2:
        return "REDUCE batch=4; disable TTA; continue"

    # 3. Failsafe already triggered -> monitor only, do not intervene
    if log['failsafe_triggered']:
        if elapsed_h > 8.5:
            return "WARNING: exceeds 8.5h; switch to no-TTA student"
        return "MONITOR; failsafe handling it"

    # 4. Slow but no failsafe -> manual early switch
```

```
    expected_total = elapsed_h / max(pct_complete, 0.01)
    if expected_total > 8.5 and pct_complete < 0.75:
        return "MANUAL SWITCH to student now; continue with TTA"

    # 5. On track -> continue
    return "CONTINUE; on track"
```

14.3 Contingency Matrix

The contingency matrix below enumerates every known failure mode and the pre-planned response. The matrix is reviewed at the end of Phase 6 (M6.5) and updated if any new failure mode is discovered during the dry runs. The response column specifies the exact action; the time-to-recover column estimates how long the response takes to execute (must be ≤30 minutes to fit within the 48-hour submission buffer).

Table 23. Contingency matrix for competition-day failure modes.

Failure Mode	Detection	Response	Recovery Time
Notebook killed (OOM)	Kaggle UI shows "Error"	Reduce batch to 4; disable TTA; resubmit	20 min
Notebook killed (timeout)	Elapsed > 9 h	Switch to no-TTA student; resubmit	30 min
NaN in output CSV	Post-run validation	Restart with student-only, no ensemble	25 min
Kaggle dataset upload fails	UI error during attach	Re-upload with rsync resume; verify SHA256	15 min
Notebook stuck in queue > 2 h	UI shows "Queued"	Cancel; resubmit to alternate region	10 min
Accidental notebook overwrite	Git diff shows change	Revert to submission-branch HEAD; resubmit	15 min
Zarr dataset corrupted	Random-read returns NaN	Switch to fallback Zarr (snapshot from T−72 h)	20 min
ONNX export numerical mismatch	Equivalence test fails	Fall back to PyTorch native inference (slower)	10 min
TensorRT engine build fails	Build exception	Already handled: fall back to ONNX Runtime	0 min (auto)
Failsafe fires too early	Switch at <60% budget	Investigate post-hoc; do NOT intervene during run	0 min
Public LB much lower than CV	Score revealed	Do NOT resubmit; investigate post-hoc	0 min
Last-minute code change breaks build	Import error	Revert to last green commit; resubmit	10 min

14.4 Communication & Escalation Protocol

Both engineers are on a shared video call from T−4 hours through submission acceptance. A shared Slack/Discord channel logs every action with a timestamp. Any decision that deviates from the playbook (e.g., a manual failsafe override) requires explicit verbal confirmation from both engineers and is logged in

the channel with the rationale. If the engineers disagree on a response, the conservative option (preserve a valid submission) is chosen by default. The escalation protocol is: (1) engineer-on-call proposes an action; (2) second engineer verifies the action matches the contingency matrix; (3) if no match, both engineers must agree on the deviation; (4) action is executed and logged. This protocol eliminates the single-point-of-failure risk where one engineer makes a fatigued decision at hour 7 of the submission run.

15. Pre-Submission Validation Protocol

The pre-submission validation protocol is the final gate between a working pipeline and a submitted pipeline. It consists of five layers: (1) sanity tests, (2) numerical equivalence verification, (3) format validation, (4) a smoke test on the public LB, and (5) a final Go/No-Go gate. The protocol is fully automated — every test is a pytest test that returns green or red — and the Go/No-Go gate is a single script that prints "GO" only if all tests are green. No submission is made until the gate prints GO. The protocol is executed at the end of Phase 6 (M6.1) and again immediately before the T−48-hour submission (T−72 h checkpoint).

15.1 Sanity Test Suite

The sanity test suite verifies the basic correctness of the pipeline: the DICOM decoder produces non-NaN arrays; the Zarr store is readable; the model forward pass produces finite outputs; the loss function returns a scalar; the gradient backprop completes without error; and the inference loop produces a CSV with the expected shape. These tests are fast (under 60 seconds total) and catch the most common integration bugs. The tests are run on every commit to the submission branch via CI; a red test blocks the commit.

[python]

```
# tests/test_sanity.py
import pytest
import torch
import numpy as np

def test_decoder_no_nan(sample_dcm_path):
    arr = decode_one(sample_dcm_path)
    assert not np.isnan(arr).any(), "Decoder produced NaN"
    assert arr.dtype == np.float16

def test_zarr_readable(zarr_path):
    import zarr
    z = zarr.open(zarr_path, mode='r')
    arr = z['volumes'][0]
    assert arr.shape == (48, 256, 256)
    assert not np.isnan(arr).any()

def test_model_forward_finite(model, dummy_batch):
    out = model(dummy_batch)
    assert torch.isfinite(out).all(), "Model output contains NaN/Inf"
    assert out.shape == (1, 12)

def test_loss_scalar(model, dummy_batch):
```

```
    out = model(dummy_batch['image'])
    loss = asymmetric_loss(out, dummy_batch['label'], dummy_batch['mask'])
    assert loss.dim() == 0, "Loss is not scalar"
    assert torch.isfinite(loss)

def test_inference_csv_shape(inference_csv, n_test):
    import pandas as pd
    df = pd.read_csv(inference_csv)
    assert df.shape == (n_test, 13), f"Expected ({n_test}, 13), got {df.shape}"
    assert (df.iloc[:, 1:] >= 0).all().all(), "Probabilities < 0"
    assert (df.iloc[:, 1:] <= 1).all().all(), "Probabilities > 1"
```

15.2 Numerical Equivalence Verification

The numerical equivalence test verifies that the ONNX/TensorRT model produces the same outputs as the PyTorch model, within a tolerance of 1e-4. This catches the silent regressions that occur when ONNX export drops an operator, quantises a layer unexpectedly, or applies a different padding convention. The test is run on 100 random inputs; the maximum absolute difference between PyTorch and ONNX outputs must be < 1e-4. If the test fails, the export is debugged (usually a dynamic axes issue or an unsupported operator) or the code falls back to PyTorch native inference (slower but correct). This test is the single most effective defence against the "ONNX AUC < PyTorch AUC" failure mode (risk R12).

[python]

```
# tests/test_numerical_equivalence.py
import torch
import numpy as np
import onnxruntime as ort

def test_onnx_matches_pytorch(pytorch_model, onnx_path, n_samples=100):
    pytorch_model.eval()
    sess = ort.InferenceSession(onnx_path, providers=['CUDAExecutionProvider'])
    max_diff = 0.0
    with torch.no_grad():
        for _ in range(n_samples):
            x = torch.randn(1, 1, 48, 256, 256, device='cuda')
            pt_out = torch.sigmoid(pytorch_model(x)).cpu().numpy()
            onnx_out = sess.run(None, {'input': x.cpu().numpy()})[0]
            onnx_out = 1 / (1 + np.exp(-onnx_out))  # sigmoid
            diff = np.abs(pt_out - onnx_out).max()
            max_diff = max(max_diff, diff)
    assert max_diff < 1e-4, f"ONNX/PyTorch mismatch: max diff = {max_diff}"
    print(f"Numerical equivalence verified: max diff = {max_diff:.2e}")
```

15.3 Format Validation

The format validation test verifies that the submission CSV matches the exact format expected by Kaggle: the column names match the sample submission; the row count matches the test set size; all probability values are in [0, 1]; and there are no duplicate study IDs. This test catches the late-rejection failure mode

(risk R14) where Kaggle accepts the notebook but rejects the output after the run completes, which can consume hours of the submission window. The test is run on the dry-run output (M6.3, M6.4) and again on the actual submission output before the run is considered complete.

[python]

```
# tests/test_format.py
import pandas as pd
import numpy as np

def test_submission_format(sub_csv, sample_sub_csv, n_test):
    sub = pd.read_csv(sub_csv)
    sample = pd.read_csv(sample_sub_csv)
    # Column names match
    assert list(sub.columns) == list(sample.columns), \
        f"Columns mismatch: {list(sub.columns)} vs {list(sample.columns)}"
    # Row count
    assert len(sub) == n_test, f"Row count: {len(sub)} vs {n_test}"
    # Probabilities in [0, 1]
    probs = sub.iloc[:, 1:].values
    assert (probs >= 0).all() and (probs <= 1).all(), "Probs out of [0,1]"
    # No duplicate study IDs
    assert sub.iloc[:, 0].nunique() == n_test, "Duplicate study IDs"
    # Study IDs match sample (same set)
    assert set(sub.iloc[:, 0]) == set(sample.iloc[:, 0]), "Study ID set mismatch"
    print("Format validation passed.")
```

15.4 Smoke Test on Public LB

The smoke test submits a 10-study inference run to the public LB to verify end-to-end notebook execution on the Kaggle infrastructure. This is not a competitive submission — the 10-study output will score poorly — but it verifies that the notebook runs on Kaggle's T4 instances, that the attached Zarr dataset is accessible, that the ONNX/TensorRT engine builds, and that the output CSV is accepted by the Kaggle submission system. The smoke test is run at T−72 hours (during Phase 6) and consumes one of the daily submission quotas. If the smoke test fails, the issue is debugged before the real submission; if it passes, the team has high confidence that the real submission will execute correctly. The smoke test is the single most effective defence against the "submission notebook hangs on launch" failure mode (risk R18).

15.5 Final Go/No-Go Gate

The final Go/No-Go gate is a single script (go_no_go.py) that runs all five validation layers and prints "GO" or "NO-GO" with a list of failed tests. The gate is run at T−72 hours and again at T−48 hours immediately before the real submission. No submission is made unless the gate prints GO. If the gate prints NO-GO at T−48 hours, the team executes the contingency matrix (Section 14.3) for the failing test; if the contingency cannot be resolved within 4 hours, the team falls back to the previous green gate (the last commit on which the gate printed GO) and submits that. The gate is the single source of truth for submission readiness; no engineer judgement overrides a NO-GO.

[python]

```python
# go_no_go.py
import subprocess
import sys

def run_gate():
    tests = [
        ("tests/test_sanity.py", "Sanity"),
        ("tests/test_numerical_equivalence.py", "Numerical equivalence"),
        ("tests/test_format.py", "Format validation"),
        ("tests/test_risk_register.py", "R1-R18 risk register"),
    ]
    failures = []
    for test_file, label in tests:
        result = subprocess.run(
            [sys.executable, "-m", "pytest", test_file, "-v"],
            capture_output=True, text=True,
        )
        status = "GREEN" if result.returncode == 0 else "RED"
        print(f"[{status}] {label}")
        if result.returncode != 0:
            failures.append((label, result.stdout))
    # Smoke test (manual verification on Kaggle UI)
    smoke_passed = input("Did the 10-study smoke test pass on Kaggle? (y/n): ").strip().lower() == 'y'
    if not smoke_passed:
        failures.append(("Smoke test", "Manual check failed"))
    if not failures:
        print("\n=== GO ===")
        print("All validation layers green. Proceed with submission.")
    else:
        print("\n=== NO-GO ===")
        print("Failing layers:")
        for label, _ in failures:
            print(f"  - {label}")
        print("Do NOT submit. Resolve failures or fall back to last green gate.")
        sys.exit(1)

if __name__ == "__main__":
    run_gate()
```

✓ The Cardinal Rule

A submitted 0.93 AUC beats an unsubmitted 0.96 AUC. The Go/No-Go gate exists to enforce this rule: when in doubt, submit the last known-good configuration rather than risk a missed deadline chasing a marginal improvement. Every Kaggle grandmaster has lost at least one competition to the "one more experiment" trap; the gate is what prevents it.

16. References

The following sources support the key technical claims in this document. Sources are tiered by credibility per the citation module: ★★★ official / peer-reviewed; ★★☆ reputable secondary; ★☆☆ general web (context only). v2.0 adds references [16]–[23] covering the new modules in Sections 11–15 (SAM, SWA, EMA, MixUp/CutMix, Vision Transformers for knee MRI, pseudo-labelling, ONNX/TensorRT, and the RSNA 2026 challenge announcement).

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